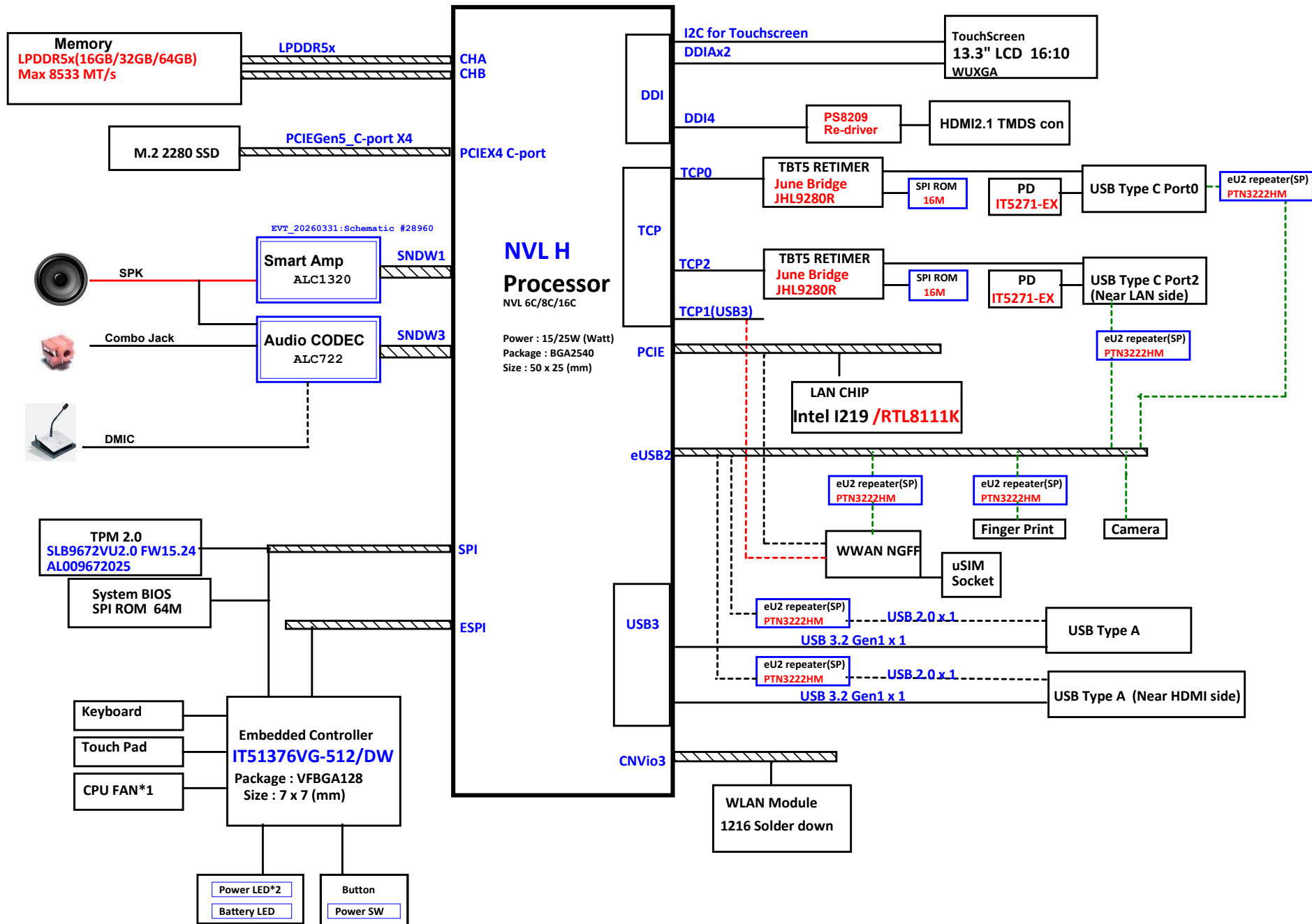


Intel Nova Lake Block Diagram





OS status									
H/W status	S0/MO3	C10	S0ix/M-off	S3/M3	S3/M-off	S4/S5/M3	S4/S5/M-off	PseudoG3	G3
RUN_ON(SLP_S3#)	H	H	H	L	L	L	L	L	L
+5V	H	H	H	L	L	L	L	L	L
+3V	H	H	H	L	L	L	L	L	L
+1.8V	H	H	H	L	L	L	L	L	L
+1.2V	H	H	H	L	L	L	L	L	L
+VDDQ	H	H	H	L	L	L	L	L	L
SUS_ON(SLP_S4#)	H	H	H	H	H	L	L	L	L
+VDD2	H	H	H	H	H	L	L	L	L
S5_ON(PRIM_VR_EN)	H	H	H	H	H	H	H	L	L
+3V_S5	H	H	H	H	H	H	H	L	L
+VCCPRIM_VNNAON	H	H	H	H	H	H	H	L	L
+VCCPRIM_VNNAON_FLTR	H	H	H	H	H	H	H	L	L
+VCCPRIM_IO	H	H	H	H	H	H	H	L	L
+VCCPRIM_IO_FLTR	H	H	H	H	H	H	H	L	L
+1.8V_S5(+VCCPRIM_PROC)	H	H	H	H	H	H	H	L	L
+VCCPRIM_PROC_FLTR	H	H	H	H	H	H	H	L	L
IMVP_VRON_EC (VR_ON)	H	L	L	L	L	L	L	L	L
+VCC_CORE	H	0V	0V	L	L	L	L	L	L
+VCC_GT	H	0V	0V	L	L	L	L	L	L
+VCCSA	H	0V	0V	L	L	L	L	L	L
+VCC_LP_ECORE	H	0V	0V	L	L	L	L	L	L

This part should not contain any substances which are specified in EM-S303.



Quanta Computer Inc.

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Size	Document Number	Rev 1A
POWER MAP		
Date:	Monday, May 04, 2026	Sheet 4 of 59



TOP side

FingerPrinter	High (R180)	Low (FP)
GPI_FP_PRS#	Absence	Present



TOP side

	High (R199)	Low (R1372)
GPI_TPM_PRS#	iTPM	dTPM



TOP side

MBID		EVT	DVT	PVT	MP
		220K (CS42201FE00)	220K (CS42201FE00)	150K (CS41501FE06)	150K (CS41501FE06)
	KR3				
	KR2	47K (CS34701FE00)	133K (CS41331FE00)	180K (CS41801FE07)	N/A



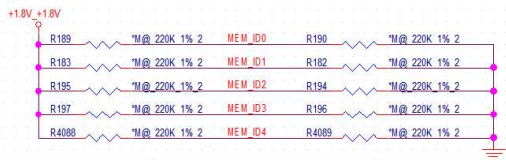
TOP side



LPDDR5 Mempry Board ID [3~0]

ID3	ID2	ID1	ID0	Board ID	VAIO P/N	QCI P/N	Vendor Part Number
R197/R196	R195/R194	R183/R182	R189/R190	GPIO			
0	0	0	0	64GB[S]10700	6V0088101	AKD5UWST518	K3KLALA0EM-MGCY
0	0	0	1	64GB[M]9600	6V0085301	AKD5UW0TL14	MT62F4G32D8DV-020 WT:D
0	0	1	0	64GB[H]10700	6V0087801	AKD5RZGTW19	H58G78EKBBX221N
0	0	1	1	32GB[S]10700	6V0088001	AKD5SWST554	K3KL9L90EM-MGCY
0	1	0	0	32GB[M]9600	6V0085201	AKD5TWSTL14	MT62F2G32D4DS-020 WT:D
0	1	0	1	32GB[H]10700	6V0087701	AKD5QGGTW01	H58G66EKBBX223N
0	1	1	0	16GB[S]10700	6V0087901	AKD5SWST553	K3KL8L80EM-MGCY
0	1	1	1	16GB[M]9600	6V0085101	AKD5SWDTL50	MT62F1G32D2DS-020 WT:D
1	0	0	0	16GB[H]10700	6V0087601	AKD5PGGTW02	H58G56EKBBX225N
1	0	0	1				

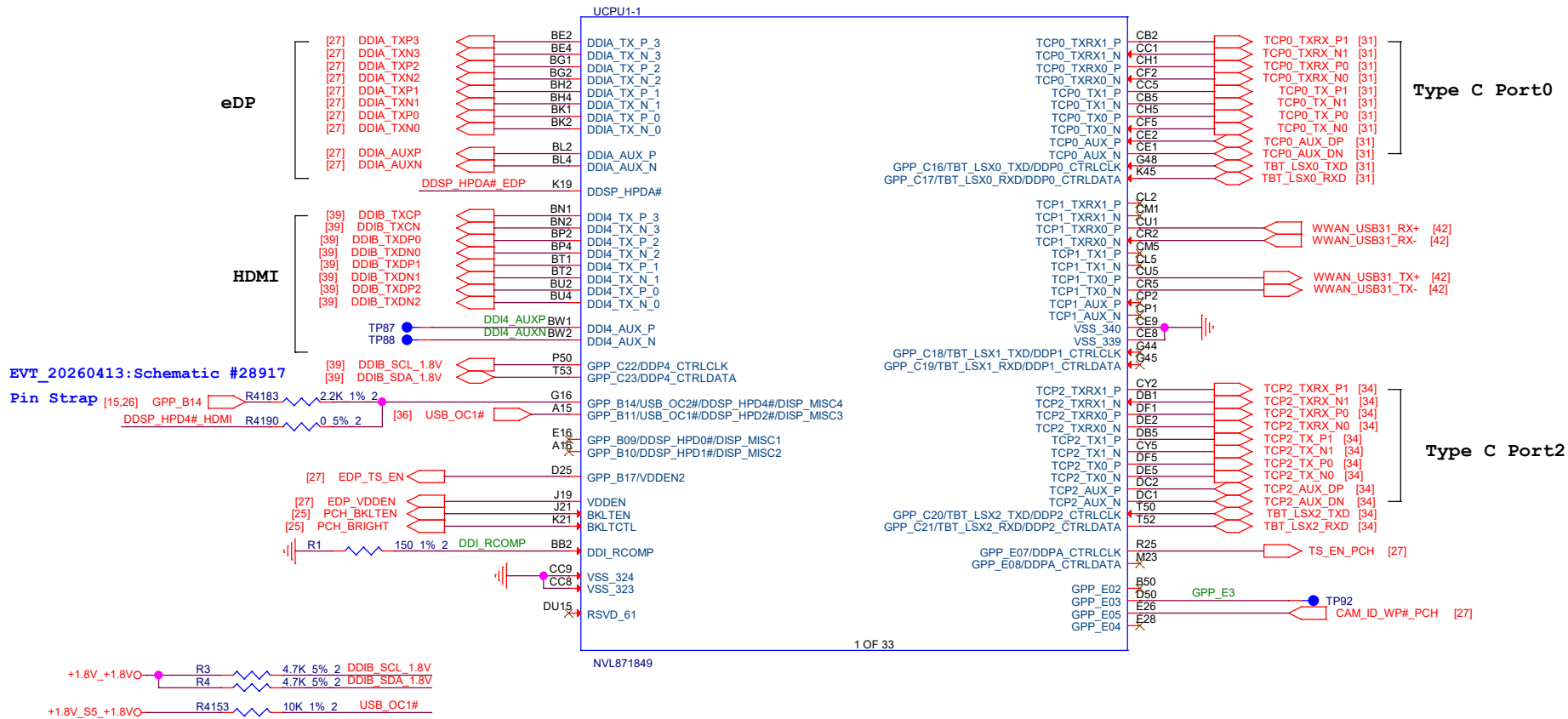
BOT side



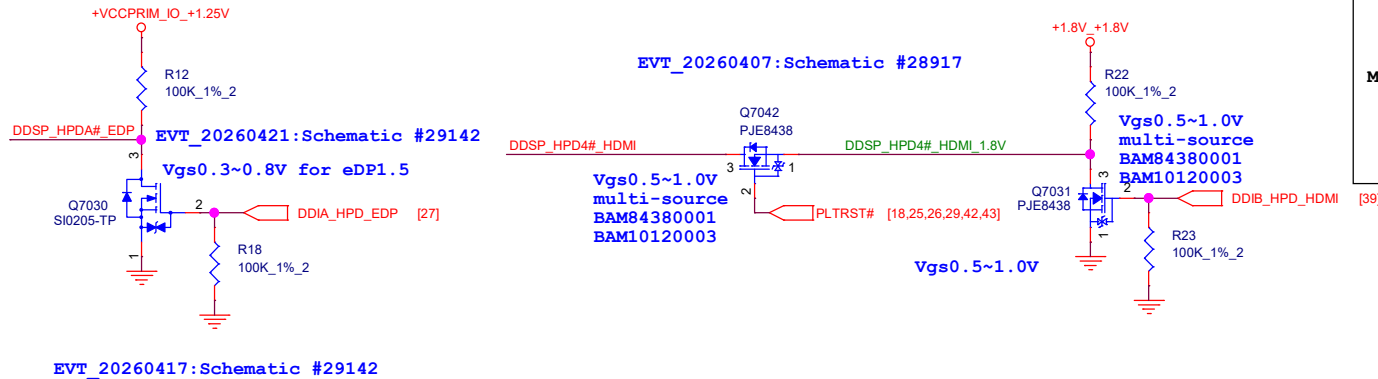
BOT side

WLAN	UWLAN1
BE211D2W (vPro) QS	ADPCBE21011
BE211D2W (N-vPro) QS	ADPCBE21012
BE213D2W (N-vPro) QS	ADPCBE21013
AX211D2W (Non-vPro)	ADPCAX21021

	MBID4_PCH	MBID5_PCH
Reserved	X	X
Please set '00' currently		
	MBID3_PCH	
w/o Vpro	0	
w/ Vpro	1	
	MBID1_PCH	MBID2_PCH
AX211	0	0
BE211	0	1
BE213	1	0
Reserve	1	1
	MBID0_PCH	
TDP15W	0	
TDP25W	1	



HPD LS

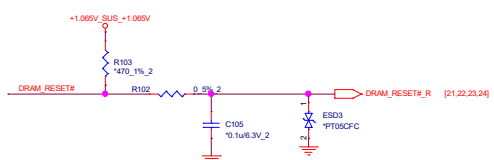


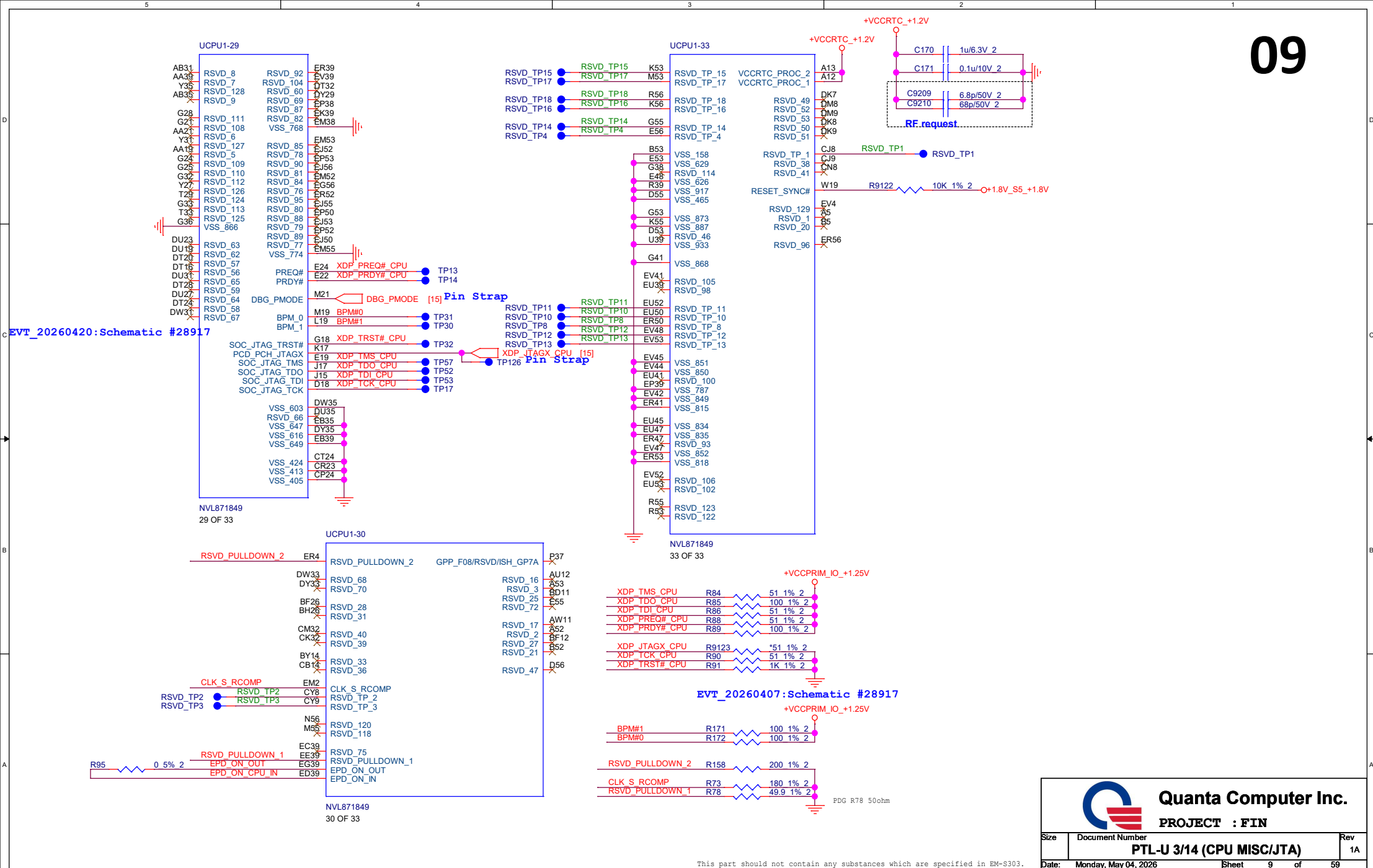
MAIN LINK (Tx)	TCP	HDMI (Default non reversed)
	TCPx_TX0_P/N	TMDS/FRL Data2
	TCPx_TX1_P/N	TMDS/FRL Data0
	TCPx_TXRX0_P/N	TMDS/FRL Data1
	TCPx_TXRX1_P/N	TMDS CLK/FRL Data3

Quanta Computer Inc.

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Size	Document Number	Rev
	PTL-U 1/14 (eDP/TCP)	1A
Date:	Monday, May 04, 2026	Sheet 7 of 59



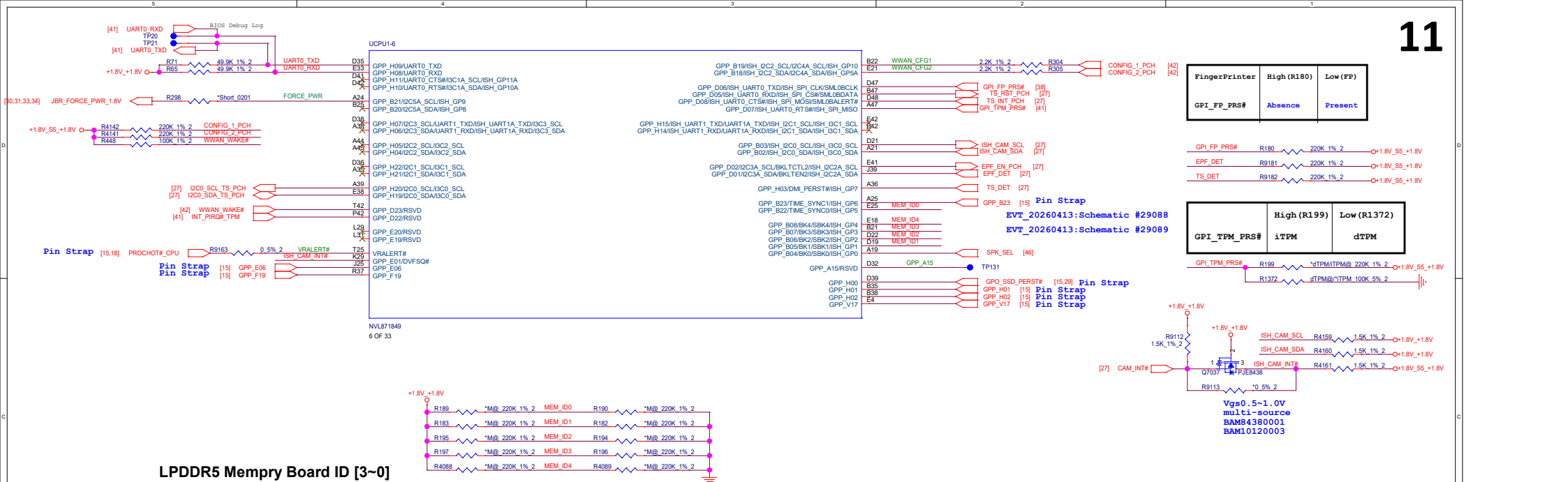


Quanta Computer Inc.

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Size	Document Number	Rev
	PTL-U 3/14 (CPU MISC/JTA)	1A
Date:	Monday, May 04, 2026	Sheet 9 of 59

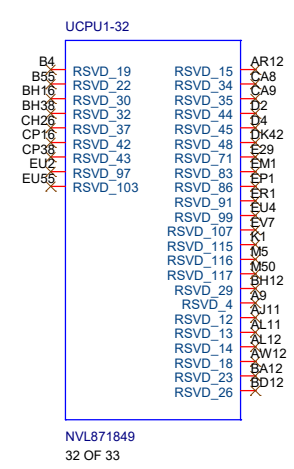
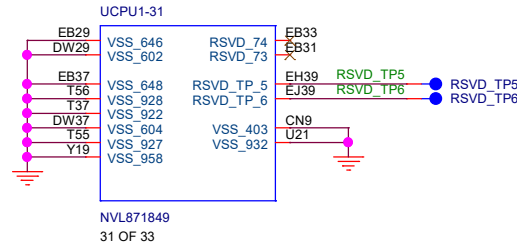
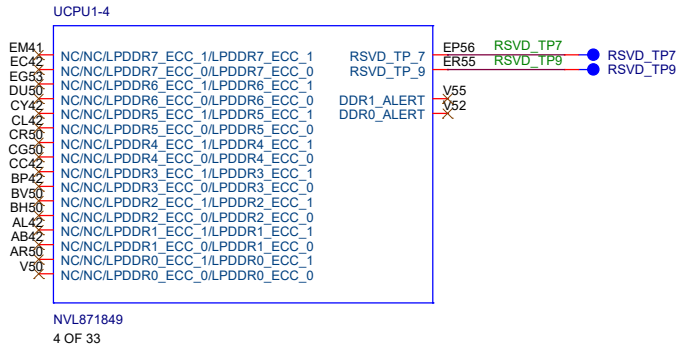




LPDDR5 Mempry Board ID [3~0]

ID3	ID2	ID1	ID0	Board ID	VAIO P/N	QCI P/N	Vendor Part Number
R197/R196	R195/R194	R183/R182	R189/R190	GPIO			
0	0	0	0	64GB[S]10700	6V0088101	AKD5UWST518	K3KLALA0EM-MGCY
0	0	0	1	64GB[M]9600	6V0085301	AKD5UW0TL14	MT62F4G32D8DV-020 WT:D
0	0	1	0	64GB[H]10700	6V0087801	AKD5RZGTW19	H58G78EKBBX221N
0	0	1	1	32GB[S]10700	6V0088001	AKD5SWST554	K3KL9L90EM-MGCY
0	1	0	0	32GB[M]9600	6V0085201	AKD5TWSTL14	MT62F2G32D4DS-020 WT:D
0	1	0	1	32GB[H]10700	6V0087701	AKD5QGGTW01	H58G66EKBBX223N
0	1	1	0	16GB[S]10700	6V0087901	AKD5SWST553	K3KL8L80EM-MGCY
0	1	1	1	16GB[M]9600	6V0085101	AKD5SWDTL50	MT62F1G32D2DS-020 WT:D
1	0	0	0	16GB[H]10700	6V0087601	AKD5PGGTW02	H58G56EKBBX225N
1	0	0	1				
1	0	1	0				
1	0	1	1				
1	1	0	0				
1	1	0	1				
1	1	1	0				
1	1	1	1	Reserve	Reserve	Reserve	Reserve

UCPU1-22										UCPU1-23										UCPU1-24										UCPU1-25										UCPU1-26										UCPU1-27										UCPU1-28											
EV50	VSS_854	VSS_793	EP7	EG23	VSS_706	VSS_631	EA11	DM11	VSS_551	VSS_486	DC5	CP5	VSS_409	VSS_329	CE11	BP46	VSS_258	VSS_191	BE55	AP52	VSS_109	VSS_41	AE53																																																
EV38	VSS_848	VSS_788	EM56	EG19	VSS_705	VSS_642	EA9	DM7	VSS_550	VSS_485	DB53	CN50	VSS_408	VSS_338	CE5	BP43	VSS_257	VSS_190	BE52	AP36	VSS_107	VSS_40	AE4	T49	VSS_925	VSS_874	G56																																												
EV35	VSS_848	VSS_788	EM56	EG19	VSS_705	VSS_642	EA8	DM5	VSS_560	VSS_485	DB53	CN50	VSS_408	VSS_338	CE5	BP40	VSS_257	VSS_190	BE5	AP32	VSS_107	VSS_40	AD55	T48	VSS_925	VSS_874	G52																																												
EV32	VSS_846	VSS_773	EM50	EG17	VSS_703	VSS_640	EA7	DM4	VSS_553	VSS_480	DB50	CN48	VSS_400	VSS_330	CE4	BP36	VSS_255	VSS_188	BE1	AP28	VSS_105	VSS_36	AD50	T45	VSS_923	VSS_871	G47																																												
EV29	VSS_845	VSS_771	EM45	EG15	VSS_702	VSS_636	EA5	DL53	VSS_550	VSS_479	DB48	CN45	VSS_399	VSS_328	CC40	BP32	VSS_254	VSS_186	BD53	AP24	VSS_104	VSS_35	AD48	T43	VSS_923	VSS_871	G47																																												
EV26	VSS_844	VSS_770	EM39	EG13	VSS_702	VSS_633	EA4	DL5	VSS_549	VSS_478	DB42	CN11	VSS_398	VSS_327	CC20	BP24	VSS_253	VSS_185	BD46	AP20	VSS_103	VSS_34	AD45	T42	VSS_923	VSS_871	G47																																												
EV24	VSS_844	VSS_770	EM39	EG13	VSS_702	VSS_633	EA4	DL5	VSS_549	VSS_478	DB40	CN7	VSS_397	VSS_325	CC55	BP11	VSS_252	VSS_183	BD43	AP5	VSS_102	VSS_33	AD12	T31	VSS_923	VSS_871	G47																																												
EV21	VSS_843	VSS_769	EM36	EG11	VSS_718	VSS_620	DY52	DL1	VSS_548	VSS_477	DB36	CM56	VSS_402	VSS_321	CC52	BP7	VSS_251	VSS_182	BD9	AP1	VSS_108	VSS_30	AD8	T15	VSS_921	VSS_870	G42																																												
EV18	VSS_842	VSS_767	EM33	EG6	VSS_717	VSS_619	DY37	DK50	VSS_477	VSS_476	DB32	CM53	VSS_395	VSS_319	CC50	BP5	VSS_259	VSS_184	BD5	AN56	VSS_97	VSS_38	AD7	T12	VSS_919	VSS_867	G39																																												
EV15	VSS_839	VSS_765	EM31	EF33	VSS_699	VSS_615	DY31	DK49	VSS_548	VSS_473	DB28	CM4	VSS_394	VSS_318	CC46	BP4	VSS_258	VSS_184	BD4	AN53	VSS_97	VSS_38	AD4	T8	VSS_919	VSS_867	G39																																												
EV12	VSS_838	VSS_764	EM28	EF5	VSS_698	VSS_614	DY21	DK46	VSS_545	VSS_472	DB24	CM2	VSS_394	VSS_318	CC44	BP4	VSS_258	VSS_184	BD2	AN50	VSS_96	VSS_32	AD2	T7	VSS_930	VSS_865	G31																																												
EV9	VSS_838	VSS_764	EM28	EF4	VSS_698	VSS_614	DY21	DK46	VSS_545	VSS_472	DB20	CL55	VSS_392	VSS_317	CC11	BN15	VSS_249	VSS_179	BD1	AN48	VSS_95	VSS_31	AD1	T5	VSS_929	VSS_864	G26																																												
EV5	VSS_856	VSS_763	EM22	EF2	VSS_697	VSS_613	DY15	DK45	VSS_543	VSS_470	DB11	CL52	VSS_388	VSS_314	CC7	BN5	VSS_246	VSS_178	BD40	AN45	VSS_93	VSS_29	AC17	R35	VSS_926	VSS_863	G22																																												
EV2	VSS_853	VSS_762	EM19	EF1	VSS_696	VSS_612	DY5	DK43	VSS_542	VSS_469	DB9	CL50	VSS_386	VSS_314	CC4	BN4	VSS_246	VSS_177	BD36	AN42	VSS_91	VSS_27	AC15	R27	VSS_916	VSS_862	G22																																												
UCPU6	VSS_836	VSS_760	EM16	EE50	VSS_691	VSS_611	DY1	DK40	VSS_540	VSS_484	DB8	CL46	VSS_386	VSS_316	CC4	BN4	VSS_247	VSS_177	BD36	AN42	VSS_91	VSS_27	AC15	R27	VSS_915	VSS_861	G15																																												
UC44	VSS_833	VSS_759	EM13	EE48	VSS_695	VSS_618	DY10	DK36	VSS_541	VSS_488	DB7	CL43	VSS_385	VSS_315	CC55	BM48	VSS_244	VSS_175	BD28	AN12	VSS_89	VSS_25	AB50	R25	VSS_914	VSS_860	G12																																												
UC38	VSS_832	VSS_758	EM7	EE42	VSS_689	VSS_608	DY10	DK28	VSS_538	VSS_482	DB4	CL11	VSS_384	VSS_313	CB53	BM45	VSS_243	VSS_174	BD24	AN9	VSS_88	VSS_24	AB46	R2	VSS_918	VSS_859	G10																																												
UC35	VSS_831	VSS_776	EM5	EE37	VSS_688	VSS_607	DY43	DK24	VSS_537	VSS_475	DB2	CL9	VSS_382	VSS_312	CB36	BM16	VSS_242	VSS_173	BD20	AN8	VSS_100	VSS_22	AB43	P49	VSS_911	VSS_858	G9																																												
UC32	VSS_829	VSS_757	EK37	EE33	VSS_686	VSS_605	DY42	DK20	VSS_535	VSS_457	CY55	CL8	VSS_390	VSS_308	CB16	BM14	VSS_240	VSS_171	BB16	AN7	VSS_98	VSS_19	AB37	P48	VSS_910	VSS_875	G1																																												
UC29	VSS_828	VSS_756	EK35	EE25	VSS_685	VSS_601	DY42	DK16	VSS_534	VSS_456	CY52	CL4	VSS_389	VSS_311	CB1	BM11	VSS_239	VSS_167	BA27	AN4	VSS_98	VSS_19	AB37	P43	VSS_909	VSS_857	G1																																												
UC26	VSS_828	VSS_756	EK35	EE25	VSS_685	VSS_601	DY42	DK14	VSS_533	VSS_455	CY46	CL1	VSS_383	VSS_307	CA50	BM9	VSS_238	VSS_166	BA48	AL55	VSS_90	VSS_17	AB25	P19	VSS_908	VSS_828	E50																																												
UC24	VSS_827	VSS_755	EK31	EE21	VSS_684	VSS_600	DY42	DK11	VSS_532	VSS_454	CY43	CL0	VSS_381	VSS_305	CA48	BL53	VSS_245	VSS_165	BA46	AL50	VSS_86	VSS_16	AB21	P15	VSS_907	VSS_827	E47																																												
UC21	VSS_825	VSS_754	EK29	EE15	VSS_683	VSS_598	DY41	DJ55	VSS_531	VSS_453	CY43	CL0	VSS_381	VSS_305	CA48	BL53	VSS_245	VSS_165	BA46	AL50	VSS_86	VSS_16	AB21	P15	VSS_907	VSS_827	E47																																												
UC18	VSS_824	VSS_752	EK27	EE12	VSS_681	VSS_597	DY41	DJ52	VSS_530	VSS_451	CY18	CL36	VSS_379	VSS_303	CA45	BL5	VSS_236	VSS_163	BA45	AL48	VSS_83	VSS_14	AB17	P12	VSS_905	VSS_823	E15																																												
UC15	VSS_823	VSS_751	EK23	EE9	VSS_680	VSS_596	DY41	DJ4	VSS_529	VSS_450	CY14	CL36	VSS_379	VSS_303	CA45	BL5	VSS_236	VSS_163	BA45	AL48	VSS_83	VSS_14	AB17	P12	VSS_905	VSS_823	E15																																												
UC12	VSS_823	VSS_751	EK23	EE9	VSS_680	VSS_596	DY41	DJ4	VSS_529	VSS_450	CY11	CK20	VSS_378	VSS_302	CA7	BK56	VSS_235	VSS_162	BA42	AL9	VSS_82	VSS_13	AB12	P7	VSS_904	VSS_822	E7																																												
UC9	VSS_822	VSS_750	EK21	EE8	VSS_694	VSS_595	DY39	DH53	VSS_528	VSS_449	CY7	CK18	VSS_376	VSS_306	BY55	BK53	VSS_231	VSS_161	BA11	AL5	VSS_87	VSS_12	AB11	N55	VSS_912	VSS_830	E2																																												
UC1	VSS_837	VSS_749	EK19	EE7	VSS_693	VSS_610	DY39	DH50	VSS_528	VSS_448	CY74	CK18	VSS_375	VSS_301	BY5	BK50	VSS_230	VSS_160	BA9	AL4	VSS_84	VSS_11	AB7	N53	VSS_903	VSS_824	E1																																												
EV45	VSS_817	VSS_747	EK17	EE55	VSS_679	VSS_593	DY35	DH46	VSS_521	VSS_448	CY6	CJ56	VSS_372	VSS_298	BY4	BK49	VSS_227	VSS_169	BA8	AL2	VSS_81	VSS_10	AB5	N5	VSS_901	VSS_845	D52																																												
EV42	VSS_816	VSS_746	EK18	EE53	VSS_677	VSS_592	DY32	DV4	VSS_520	VSS_447	CW52	CJ56	VSS_372	VSS_298	BY1	BK42	VSS_226	VSS_168	AY56	AK53	VSS_78	VSS_10	AA56	M49	VSS_898	VSS_856	D45																																												
EV38	VSS_814	VSS_741	EJ45	ED29	VSS_676	VSS_591	DY55	DH9	VSS_517	VSS_446	CW5	CJ48	VSS_370	VSS_297	BW53	BK42	VSS_225	VSS_147	AY55	AK36	VSS_78	VSS_10	AA53	M37	VSS_897	VSS_842	D13																																												
EV35	VSS_812	VSS_740	EJ37	ED25	VSS_675	VSS_587	DY55	DH8	VSS_517	VSS_445	CW4	CJ45	VSS_368	VSS_296	BW5	BK18	VSS_224	VSS_146	AY53	AK32	VSS_76	VSS_9	AA17	M33	VSS_896	VSS_846	D10																																												
EV32	VSS_812	VSS_738	EJ37	ED25	VSS_675	VSS_588	DY55	DH8	VSS_517	VSS_445	CW2	CJ11	VSS_368	VSS_296	BW4	BK16	VSS_222	VSS_145	AY52	AK28	VSS_76	VSS_9	AA4	M29	VSS_894	VSS_846	D9																																												
EV33	VSS_811	VSS_737	EJ33	ED21	VSS_673	VSS_584	DY55	DH7	VSS_525	VSS_443	CW2	CJ11	VSS_368	VSS_296	BW4	BK16	VSS_222	VSS_145	AY52	AK28	VSS_76	VSS_9	AA4	M29	VSS_894	VSS_846	D9																																												
EV31	VSS_810	VSS_736	EJ25	ED15	VSS_672	VSS_583	DY55	DH7	VSS_525	VSS_442	CW0	CJ5	VSS_364	VSS_294	BV46	BK11	VSS_220	VSS_143	AW50	AK20	VSS_73	VSS_7	AA1	M28	VSS_892	VSS_859	B56																																												
EV29	VSS_809	VSS_735	EJ21	ED1	VSS_678	VSS_582	DY11	DH2	VSS_519	VSS_438	CW48	CJ4	VSS_369	VSS_289	BV43	BK9	VSS_219	VSS_141	AW46	AK16	VSS_72	VSS_5	Y50	M7	VSS_900	VSS_859	B45																																												
EV28	VSS_808	VSS_734	EJ15	ED53	VSS_671	VSS_581	DY11	DH1	VSS_518	VSS_437	CW45	CJ2	VSS_368	VSS_288	BV42	BK8	VSS_234	VSS_139	AW42	AK5	VSS_71	VSS_6	Y48	M2	VSS_899	VSS_857	B17																																												
EV26	VSS_807	VSS_732	EJ12	EC50	VSS_668	VSS_589	DY11	DH0	VSS_517	VSS_436	CW36	CJ3	VSS_367	VSS_287	BV40	BK7	VSS_232	VSS_140	AW9	AK1	VSS_70	VSS_5	Y45	L21	VSS_891	VSS_855	B36																																												
EV24	VSS_806	VSS_730	EJ9	EC43	VSS_666	VSS_588	DY11	DH0	VSS_516	VSS_435	CW24	CJ24	VSS_362	VSS_286	BV32	BK4	VSS_232	VSS_140	AW4	AJ48	VSS_66	VSS_5	Y41	VSS_890	VSS_854	B28																																													
EV22	VSS_804	VSS_745	EJ8	EC37	VSS_665	VSS_586	DY11	DF40	VSS_512	VSS_432	CY11	CH24	VSS_360	VSS_284	BV28	BJ19	VSS_223	VSS_137	AW2	AJ45	VSS_65	VSS_5	Y12	VSS_886	VSS_853	B24																																													
EV21	VSS_803	VSS_744	EJ7	EC35	VSS_663	VSS_580	DY11	DF36	VSS_511	VSS_431	CY9	CH4	VSS_359	VSS_283	BV24	BH55	VSS_218	VSS_136	AW1	AJ12	VSS_64	VSS_5	Y8	VSS_885	VSS_852	B19																																													
EV19	VSS_800	VSS_742	EJ5	EC33	VSS_661	VSS_579	DY3	CH8	VSS_509	VSS_429	CY8	CH2	VSS_358	VSS_281	BV20	BH52	VSS_216	VSS_134	AV36	AJ9	VSS_63	VSS_4	Y5	VSS_884	VSS_851	B18																																													
EV16	VSS_799	VSS_739	EJ2	EC29	VSS_660	VSS_576	DR56	DF32	VSS_508	VSS_440	CY7	CH9	VSS_357	VSS_280	BV14	BH48	VSS_215	VSS_133	AV28	AJ7	VSS_68	VSS_4	Y5	VSS_882	VSS_848	B12																																													
EV15	VSS_798	VSS_733	EJ1	EC27	VSS_659	VSS_574	DR43	DF20	VSS_506	VSS_430	CY53	CG43	VSS_353	VSS_279	BV11	BH45	VSS_212	VSS_132	AV24	AH56	VSS_67	VSS_3	Y37	VSS_881	VSS_847	A50																																													
EV13	VSS_797	VSS_729	EH37	EC25	VSS_658	VSS_573	DR39	DF16	VSS_505	VSS_429	CY4	CG42	VSS_352	VSS_278	BV9	BH40	VSS_211	VSS_131	AV20	AH53	VSS_62	VSS_3	Y33	VSS_880	VSS_847	A7																																													
EV12	VSS_795	VSS_727	EH33	EC23	VSS_656	VSS_575	DR5	DF14	VSS_502	VSS_427	CY2	CG11	VSS_350	VSS_292	BV8	BH36	VSS_209	VSS_129	AV16	AH5	VSS_60	VSS_3	Y29	VSS_878	VSS_844	A4																																													
EV10	VSS_794	VSS_726	EH25	EC19	VSS_655	VSS_572	DR56	DF7	VSS_501	VSS_426	CY38	CG18	VSS_350	VSS_291	BV6	BH34	VSS_208	VSS_127	AV13	AG55	VSS_59	VSS_3	Y25	VSS_877	VSS_855	AR11																																													
EV7	VSS_820	VSS_725	EH21	EC17	VSS_653	VSS_571	DP53	DF4	VSS_515	VSS_425	CY1	CG7	VSS_356	VSS_277	BV7	BH11	VSS_207	VSS_126	AV48	AG52	VSS_57	VSS_3	Y47	VSS_889	VSS_851																																														
EV2	VSS_819	VSS_724	EH15	EC15	VSS_652	VSS_570	DP36	DF2	VSS_510	V																																																													



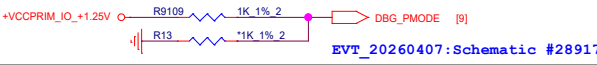
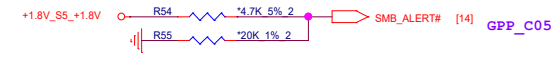
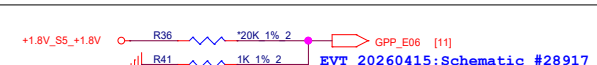
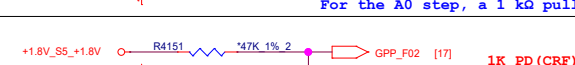
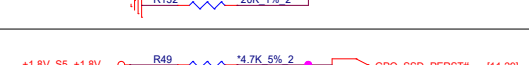
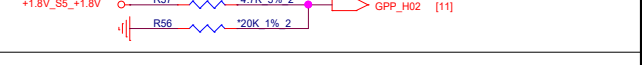
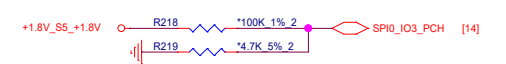
This part should not contain any substances which are specified in EM-S303.



Quanta Computer Inc.
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	PTL-H 7/14 (RSVD)	1A
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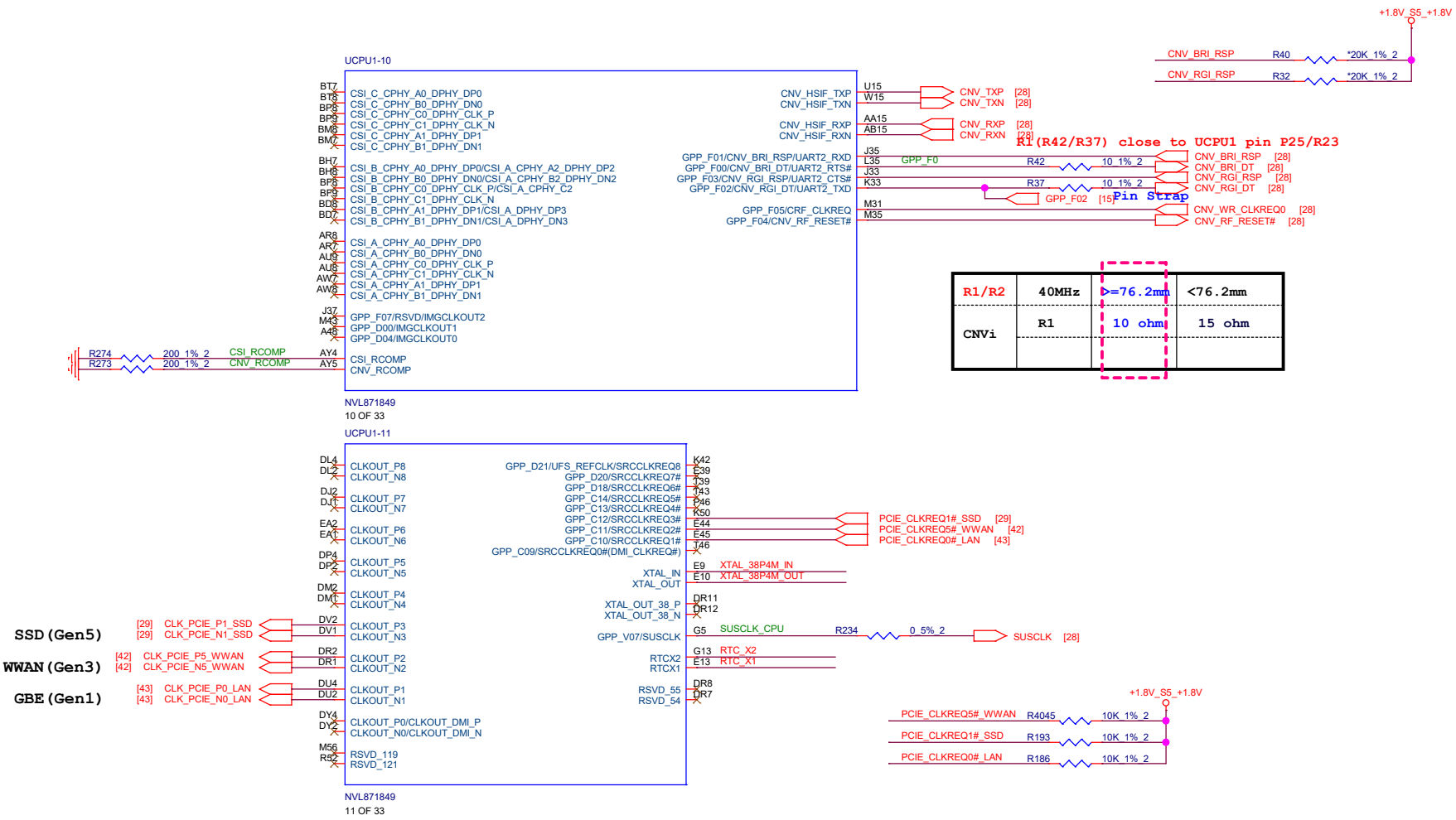


GPIO	Usage	Sampled	PU/PD (INTERNAL)	Strap
DBG_PMODE	Reserved	RSMRST#	20K PU (INT)	
GPP_B14	TOP SWAP Override 0: Disable(Default) 1: Enable	PLT_PWROK	20K PD (INT)	
GPP_B23	Reserved	RSMRST#	20K PD (INT)	
GPP_C05	eSPI Disable 0: Enable(Default) 1: Disnable	RSMRST#	20K PD (INT)	
GPP_C08	TLS CONFIDENTIALITY Enable 0: Disable(Default) 1: Enable	RSMRST#	20K PD (INT)	
GPP_C15	Reserved	RSMRST#	20K PD (INT)	
GPP_D03	Reserved	RSMRST#	20K PD (INT)	
GPP_D12	NO Reboot 0: Disable(Default) 1: Enable	PLT_PWROK	20K PD (INT)	
GPP_E06	JTAG ODT Disable 0: Disable 1: Enable(Default)	RSMRST#	20K PU (INT)	
GPP_F02	CNVi Mode Select 0: Integrated CNVi Enable 1: Integrated CNVi Disable(Default)	RSMRST#	20K PU (INT)	
GPP_F19	Reserved	RSMRST#	20K PD (INT)	
GPP_H00	eSPI Flash Sharing Mode 0: MAFS (Default) 1: SAFS	RSMRST#	20K PD (INT)	
GPP_H01	SPI Flash Descriptor Recovery 0: Disable(Default) 1: Enable	RSMRST#	20K PD (INT)	
GPP_H02	Reserved	RSMRST#	20K PD (INT)	
GPP_V17	Flash Descriptor Security Override 0: Enable(Default) 1: Disable	RSMRST#	20K PD (INT)	
PDD_PCH_JTAGX	Reserved	RSMRST#	20K PD (INT)	
SPI0_IO2	Reserved	RSMRST#	20K PU (INT)	
SPI0_IO3	Reserved	RSMRST#	20K PU (INT)	
VRALERT#	Reserved		NA	

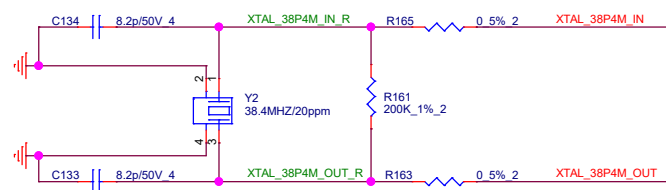
For the A0 step, a 1 kΩ pull-down is required. (2026/3/30)

For A0 step, assemble 1 kΩ Ohm PD.

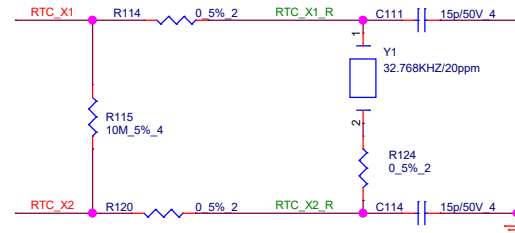




Crystal 38.4Mhz



Crystal 32.768K



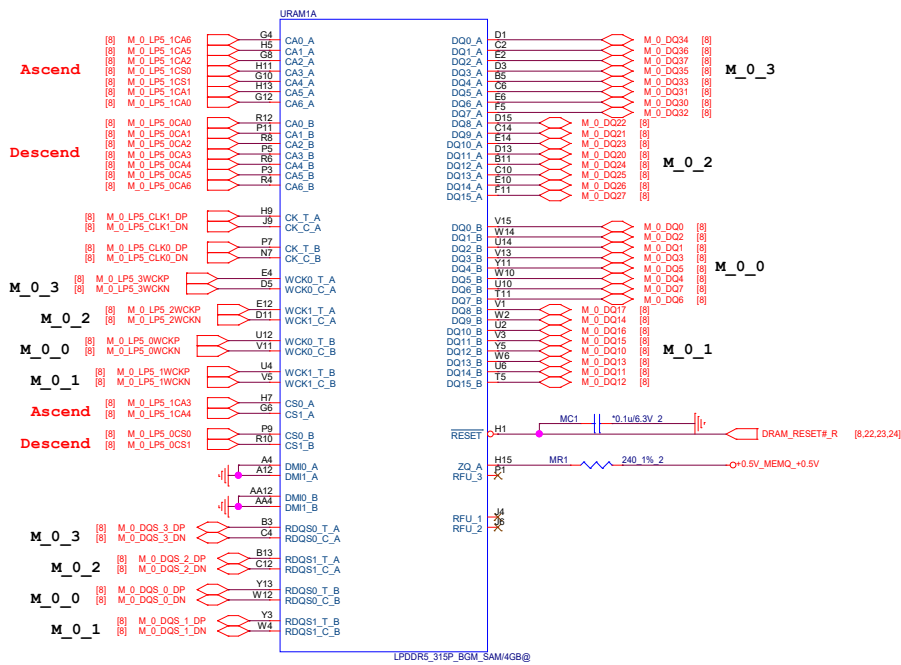




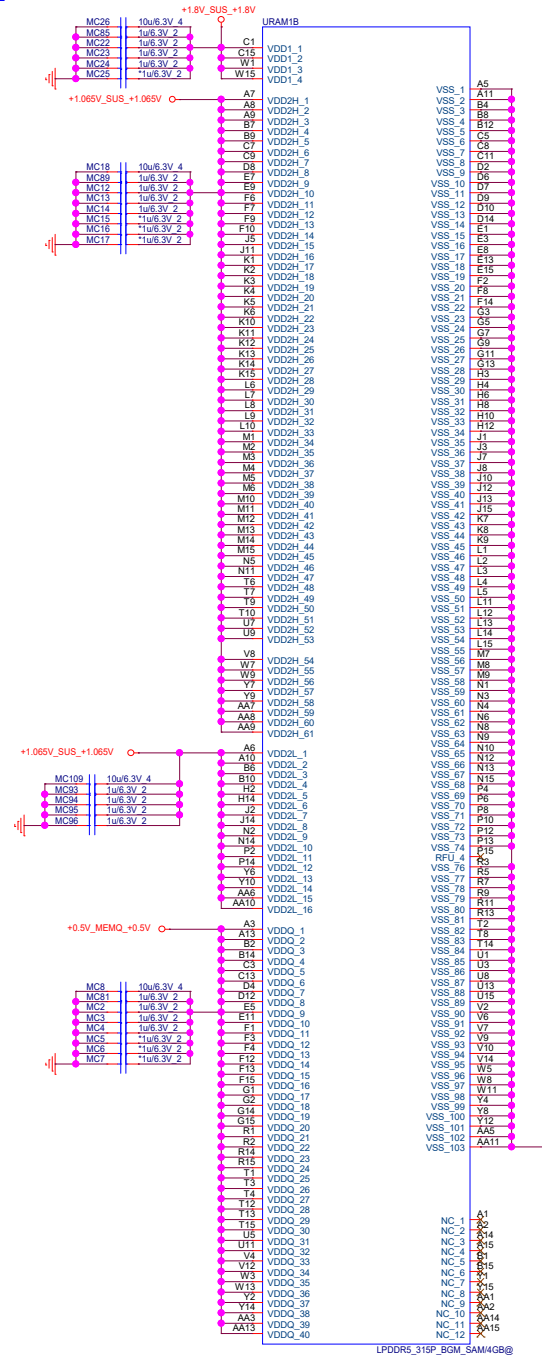
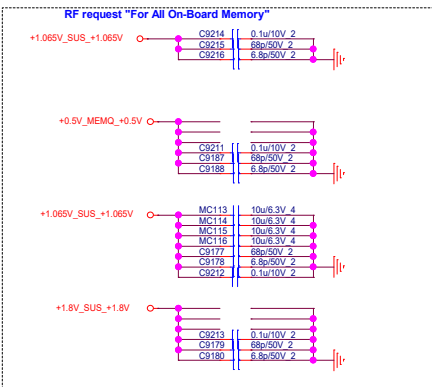


UP : VDD1>VDD2H>VDD2L>VDDQ-200mV
DOWN: VDD1>VDD2H>VDD2L>VDDQ-200mV

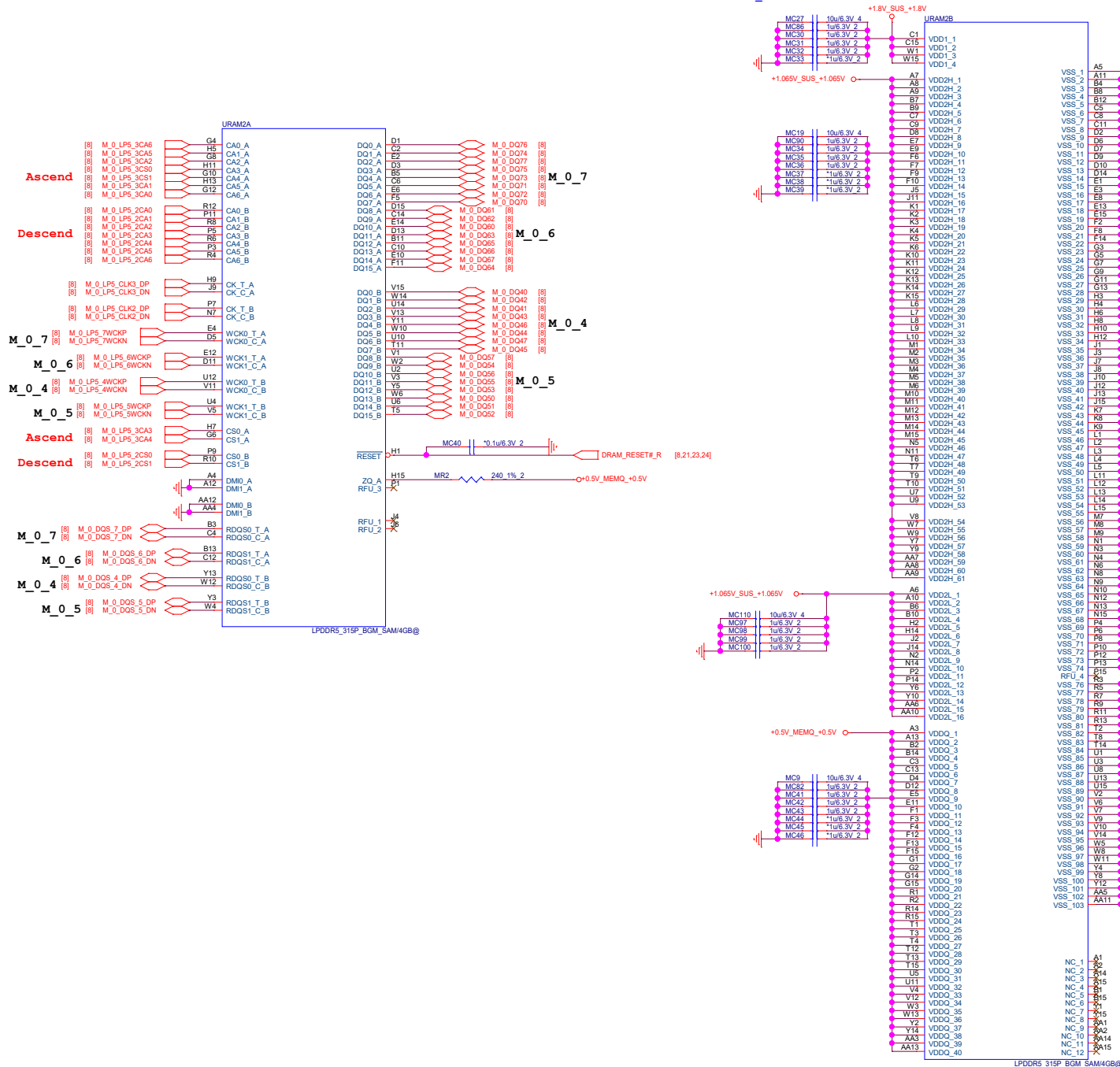
EVT_20260420:Schematic #29140

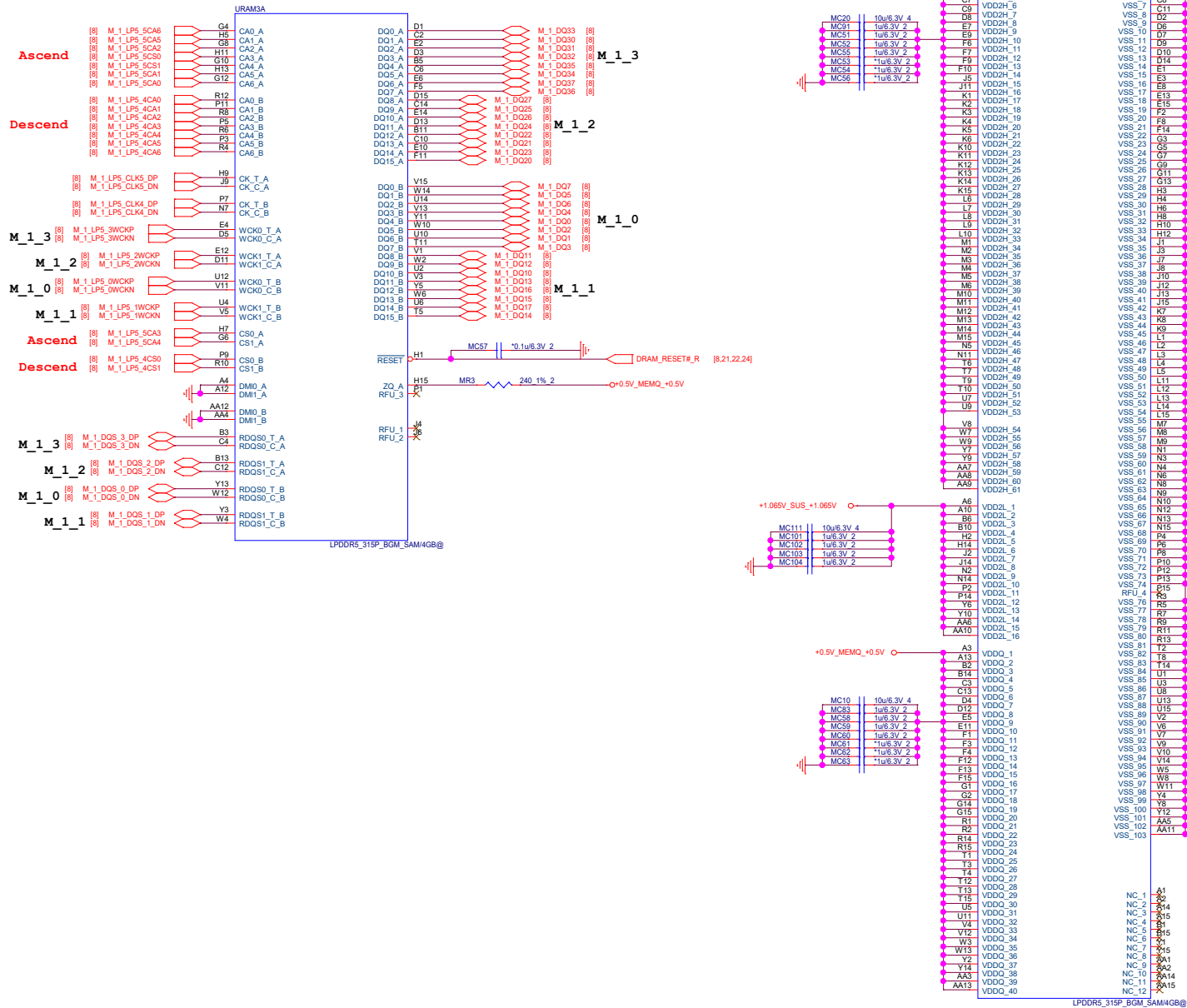


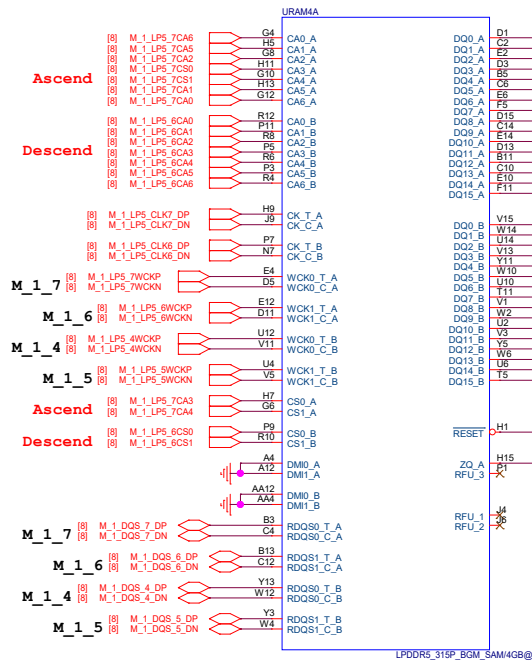
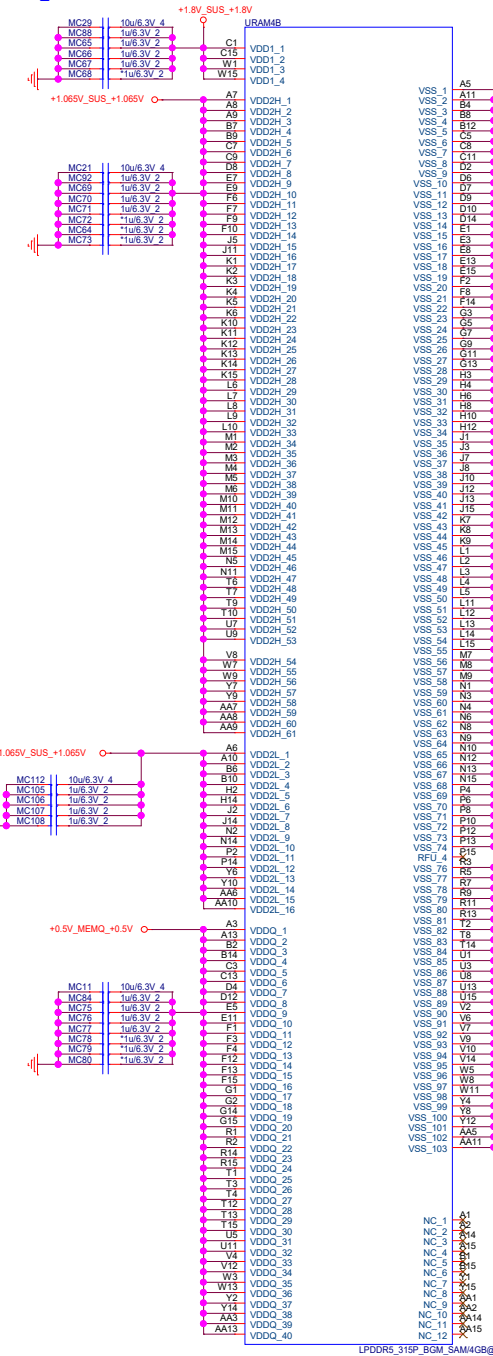
EVT_20260420:Schematic #29145

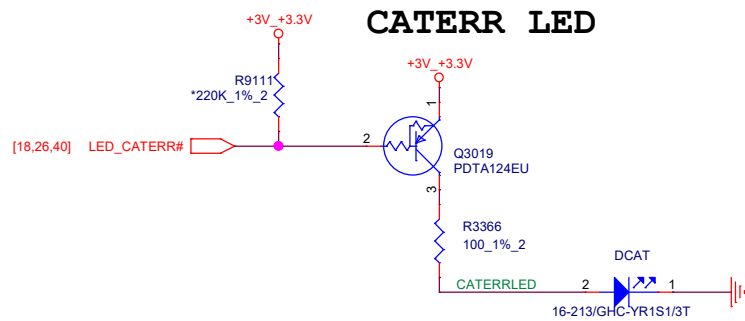
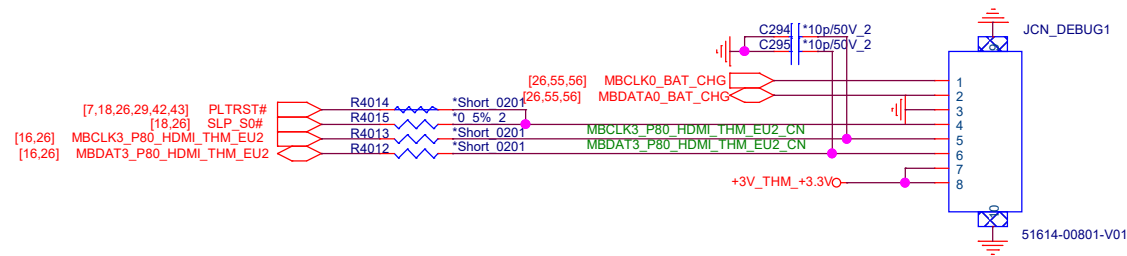
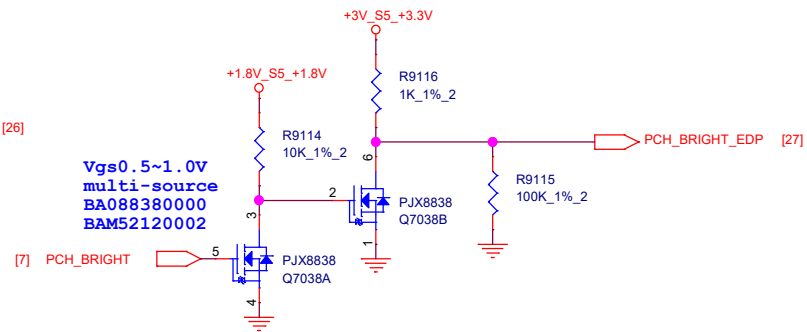
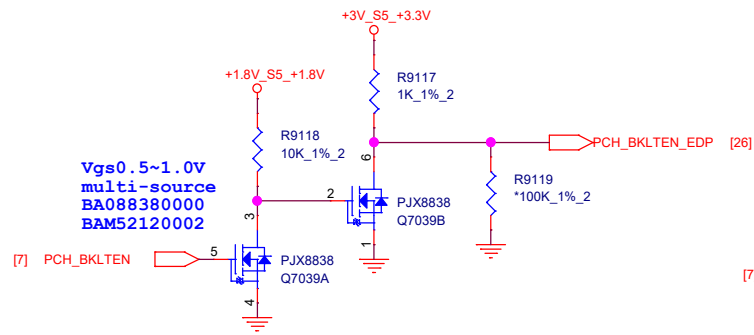


EVT_20260420:Schematic #29140

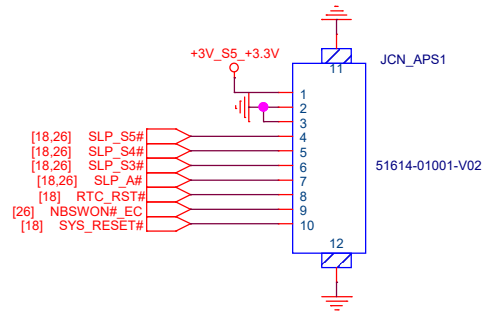








APS debug CONN



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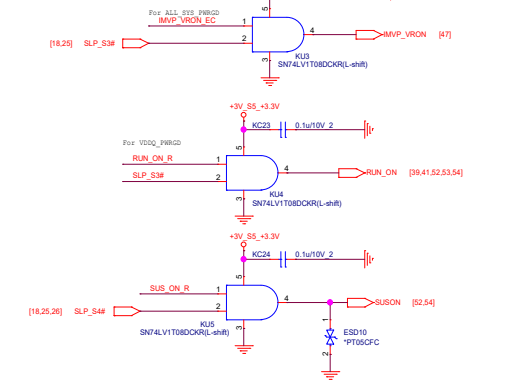
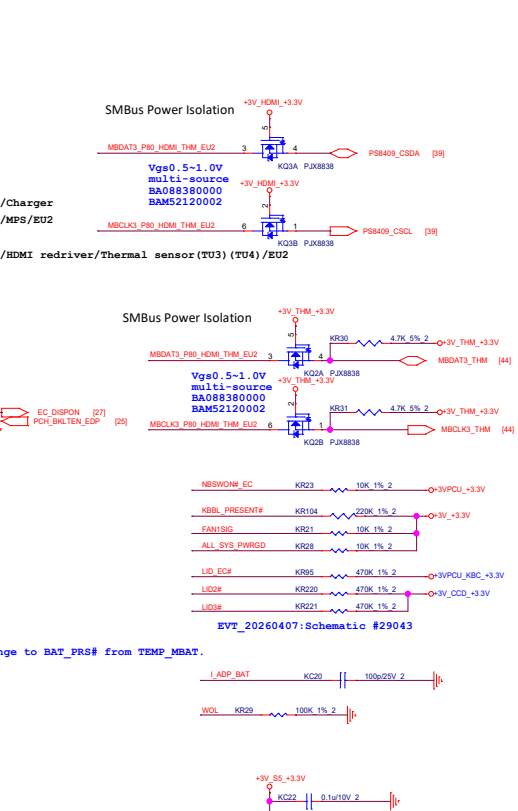
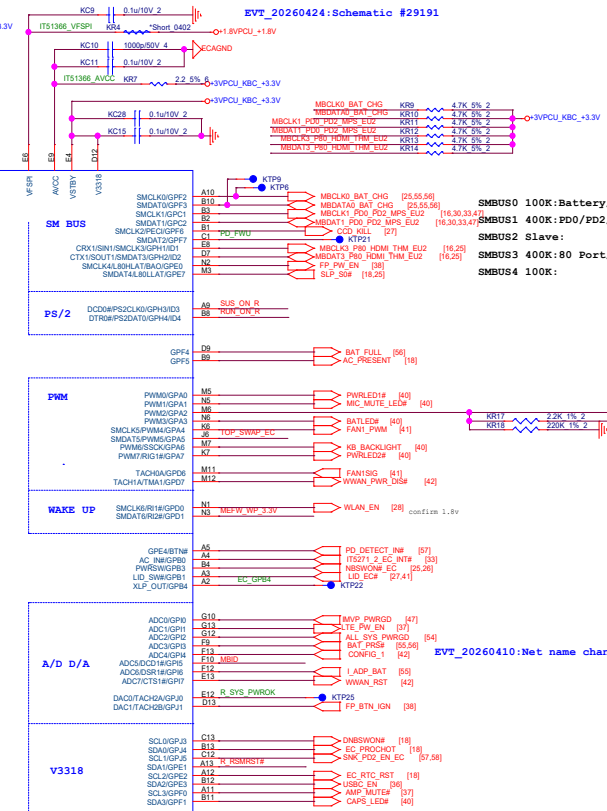
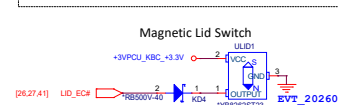
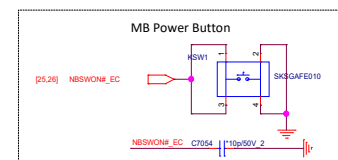
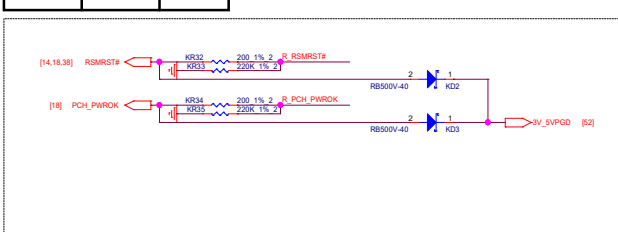
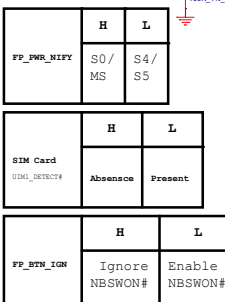
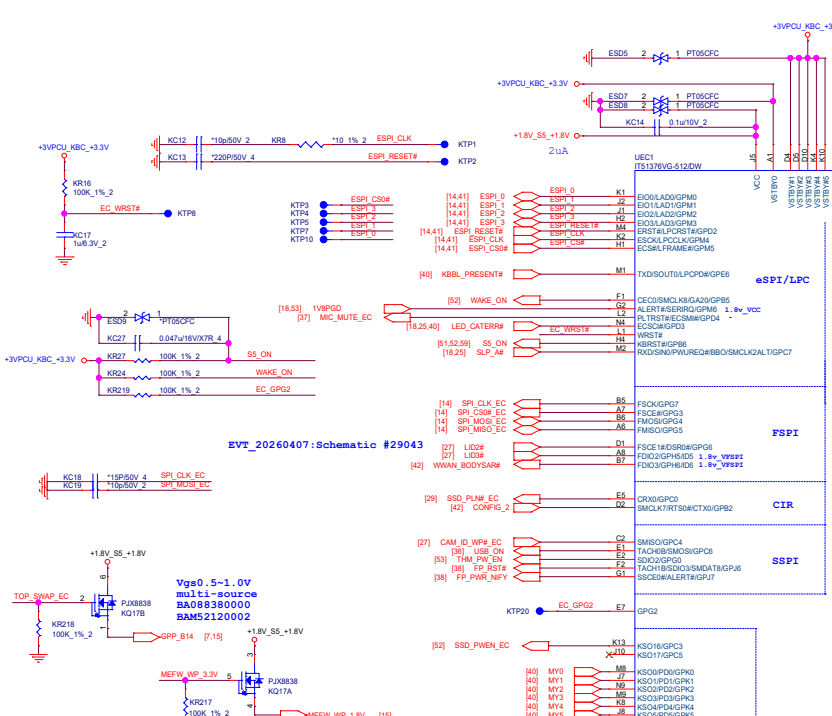
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This part should not contain any substances which are specified in EM-S303.

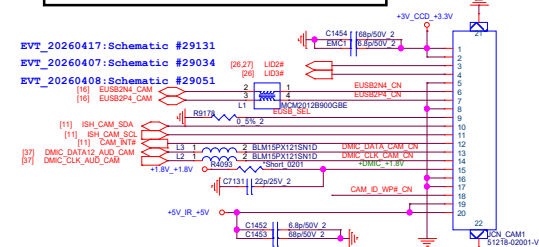
ELPM function remove

The diagram illustrates the removal of the ELPM function. It shows two circuit paths. The left path starts with a +3VDCU_+3.3V source connected to a KR1 component, which is then connected to a KTP23 component. The right path starts with a +3VDCU_KBC_+3.3V source connected to a 4, 5mA current source, which is then connected to a KBC component. The KBC component is connected to a series of resistors (KC1 to KC8) with values: 0.1u10V_2, 0.1u10V_2, 0.1u10V_2, 0.1u10V_2, 0.1u10V_2, 0.1u10V_2, 0.1u10V_2, and 0.1u10V_2. The final output is connected to a KTP24 component.

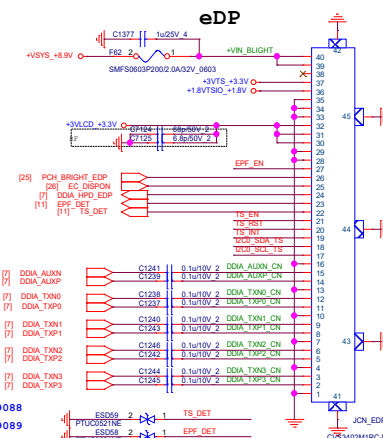
+3VPCU_KBC_+3.3V ○ 220K 1% 2 KR3 MBIO 47K 1% 2 KR2



Type	FHD Camera	4K R Camera
Vendor PN	N/A	
Resolution	1920x1080	RGB : 3840 x 2400 IR : 1280 x 720
Interface	USB2.0	USB2.0
LED	N/A	NA
Quanta PN	4MSFWCA0000	4MSFWCA0010



	Pattern3
F62 Temp. (C)	70.7

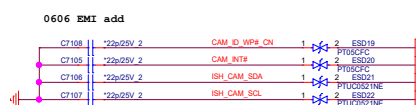


EVT_20260413:Schematic #29088
EVT_20260413:Schematic #29089

Project	NPI	MP
Fennec@JP	1.8V for verify	no-mount
13" model@QCMC		

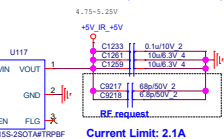
```
Vgs0.5~1.0V
multi-source
BAM84380001
RAM10120003
```

For Hello Camera



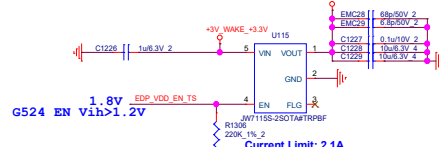
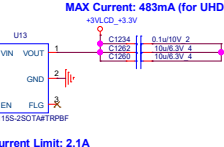
Host GPIO Direction	Host GPIO Value	Connected Camera	Result
IN	HIGH	HPD Hello cam	HPD Hello cam detected
IN	LOW	FWD Hello cam	FWD Hello cam detected
OUT	LOW	Hello cam	Write protected
OUT	HIGH	Hello cam	Write enabled

MAX Current: 740mA (for 4K IR Camera)

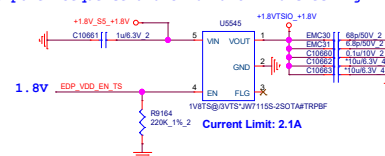


```
EVT_20260407:Schematic #29043
Vgs0.5~1.0V
multi-source
BA088380000
RAM52120002
```

Step6 Touch Screen Power

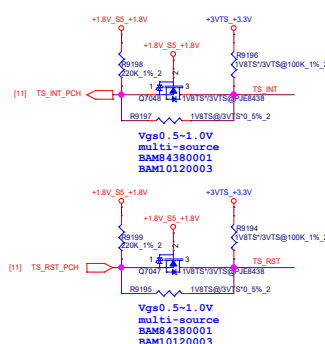
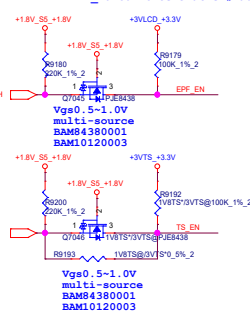


TS power sequence and GPIO wait VAI0 checking



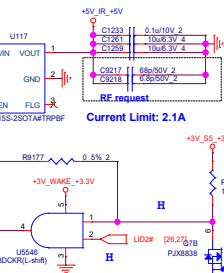
EVT_20260413:Schematic #29091 EVT_20260420:Schematic #29122
EVT_20260416:Schematic #29088

EVT 20260420:Schematic #29122



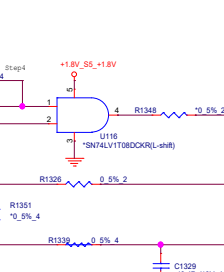
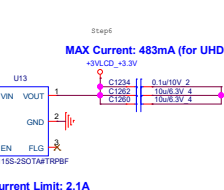
Current Limit: 800mA

4.75~5.25V

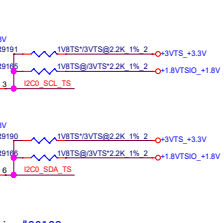


20260407: C-formation #20043

```
Vgs0.5~1.0V
multi-source
BA088380000
RAM52120002
```



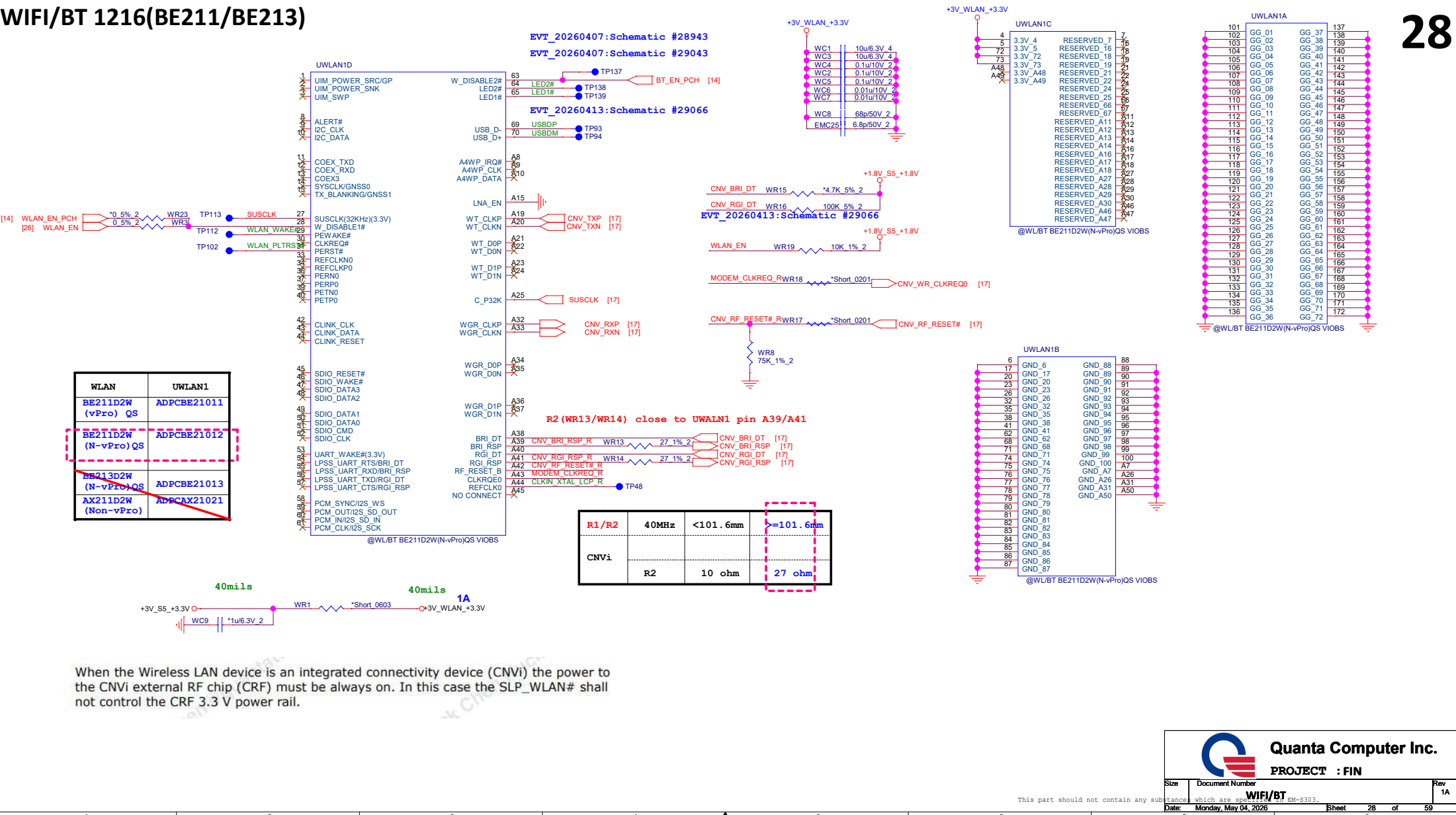
 UNIVERSITY OF MICHIGAN PRESS



0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99

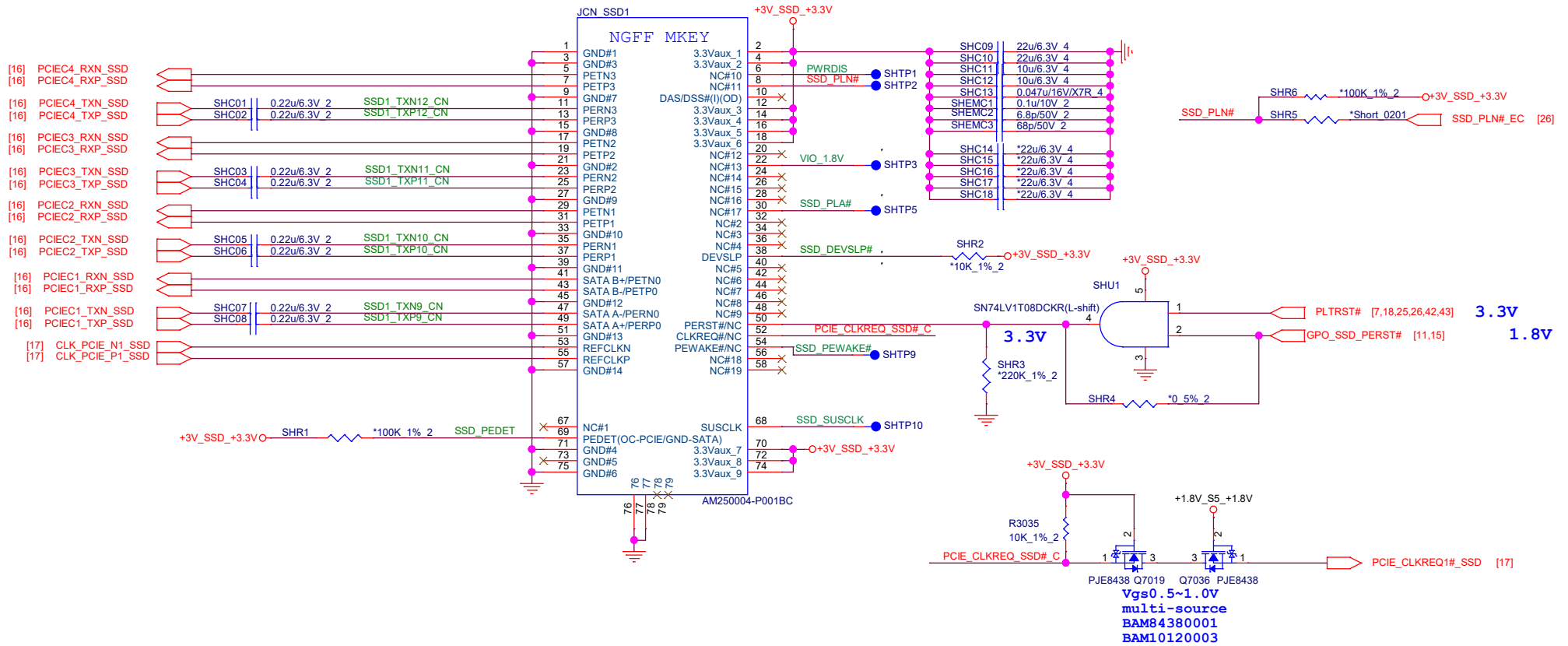
1c #29122

WIFI/BT 1216(BE211/BE213)



SSD MZVLC4T0HBL1-00B07
4TB PCIe Gen5 read=2.81A

3A



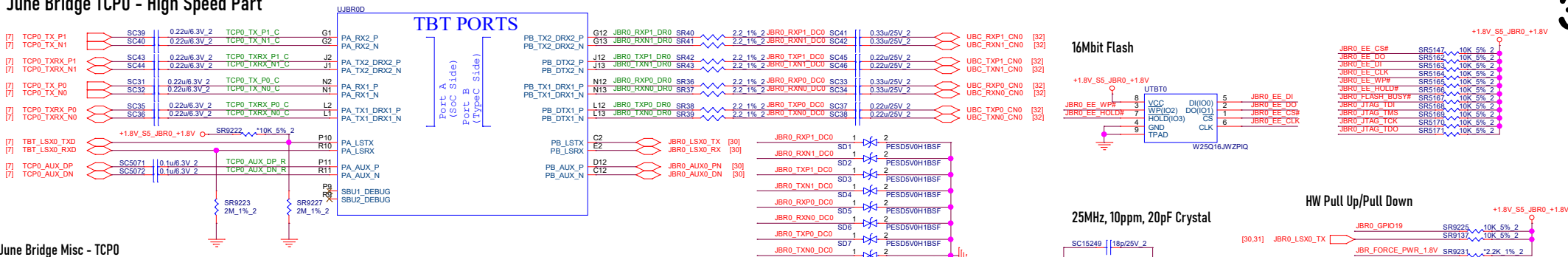
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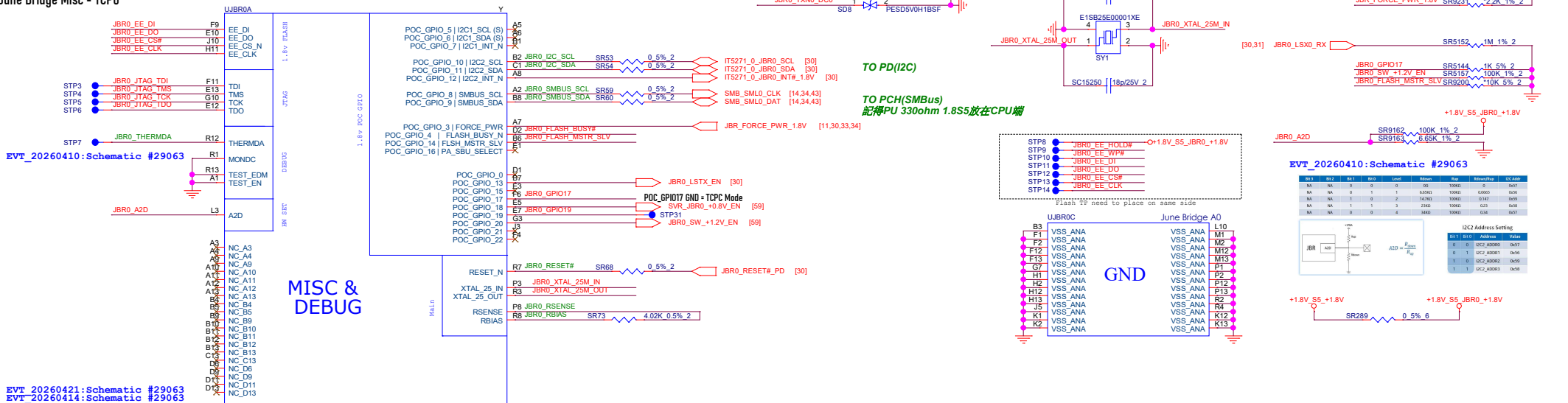
This part should not contain any substances which are specified in EM-S303.



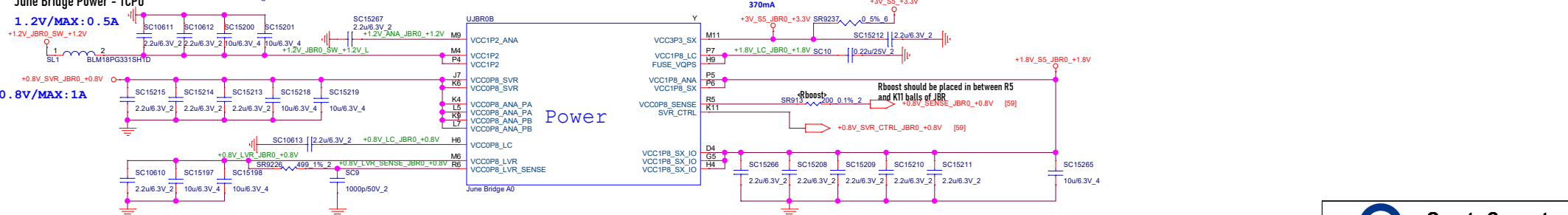
June Bridge TCP0 - High Speed Part



June Bridge Misc - TCPO

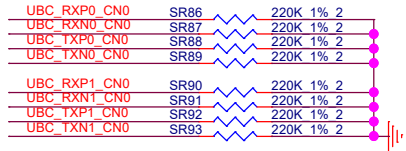


June Bridge Power - TCP0



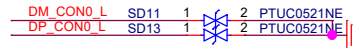
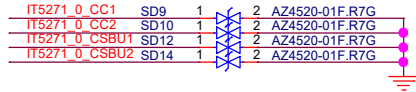
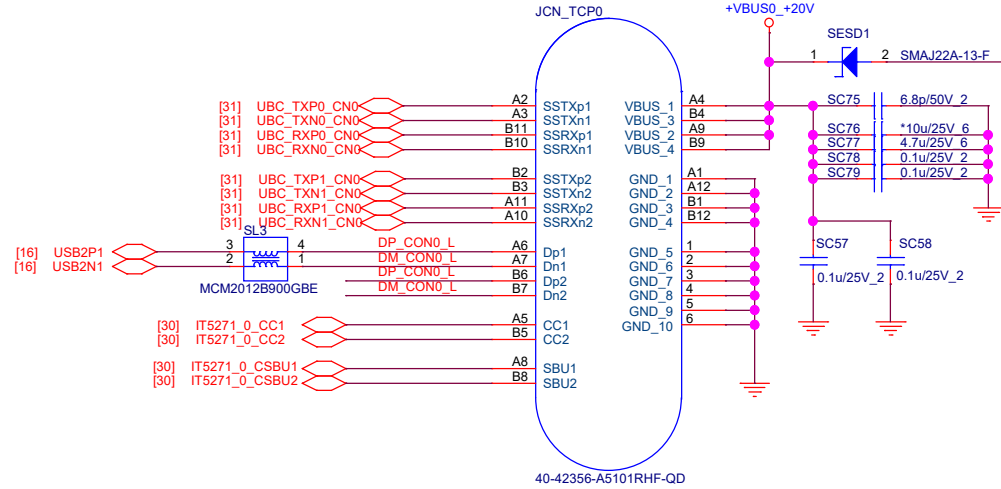
This part should not contain any substances which are specified in EM-S303.

USB Type-C Port-0

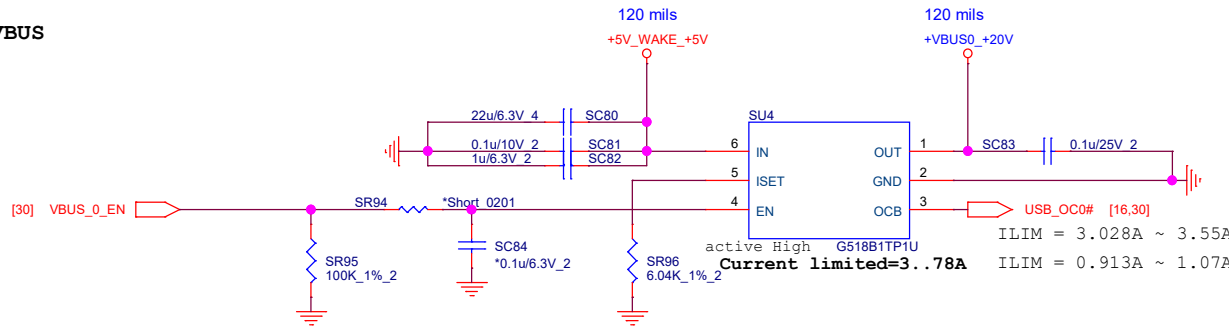


Close to CN

Bleeding SSTX/SSRX resistors - must be placed near USBC connector if 330nF cap is being used. Otherwise de-populate

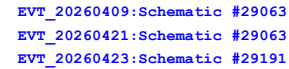


VBUS



This part should not contain any substances which are specified in EM-S303.

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Type C CON Port0		Sheet	32	of
		59		



#I2C1_ADDR	TCPC1/TCPC0 ID (default)
0V	7'h56 (8'hAC)/7'h26 (8'h4C)
Floating	7'h57 (8'hAE)/7'h27 (8'h4E)
1.8V	7'h58 (8'hB0)/7'h28 (8'h50)
3.3V	7'h59 (8'hB2)/7'h29 (8'h52)

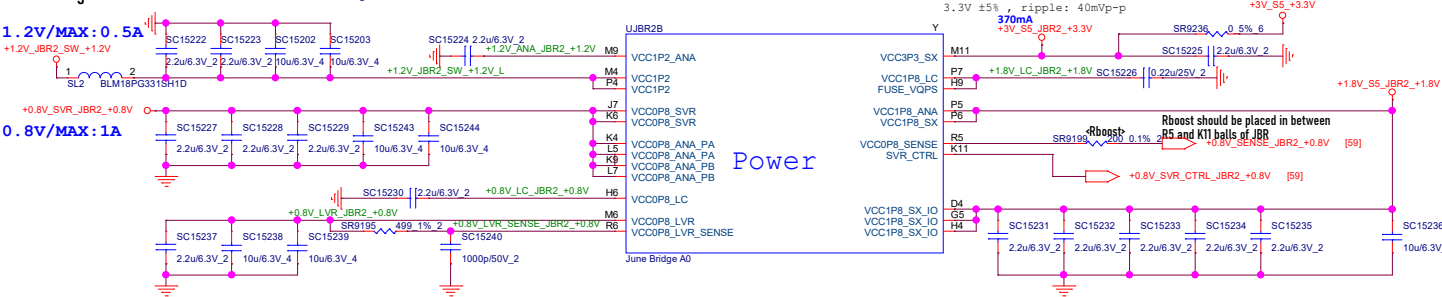
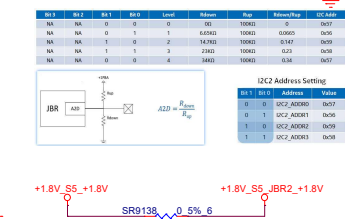
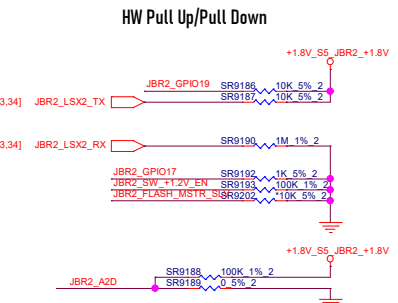
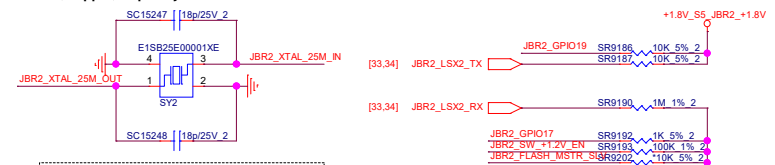
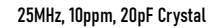
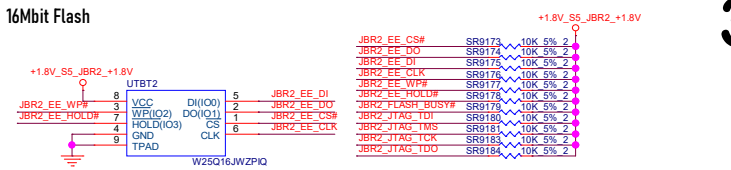
```
I2C1_slaveID decode(ADDR=floating, slave ID=7'h42
```

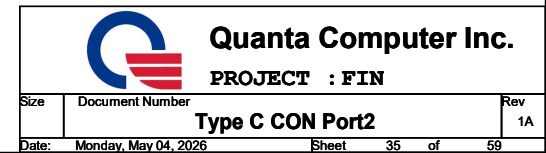


PROJECT : FIN

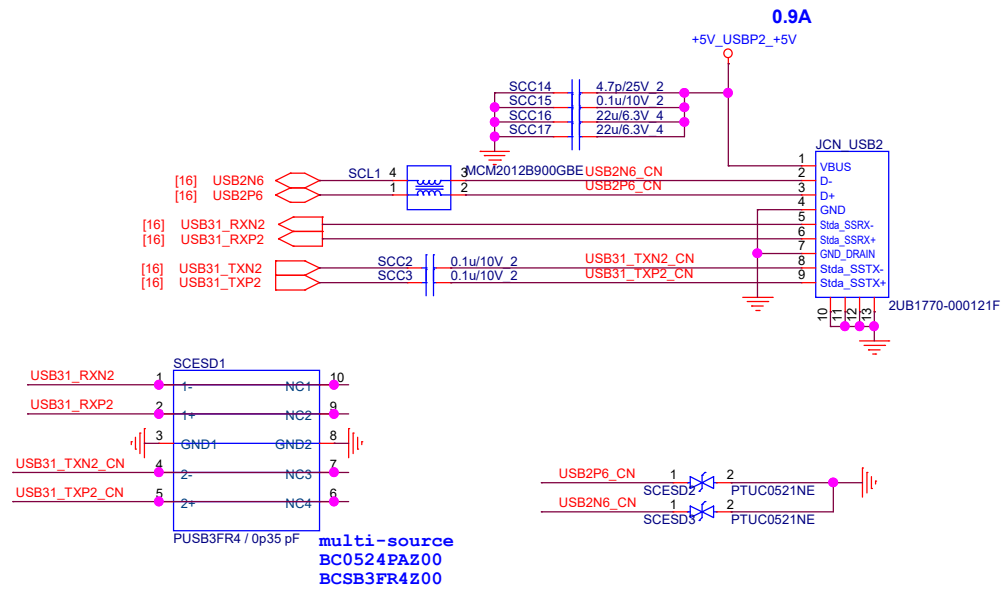
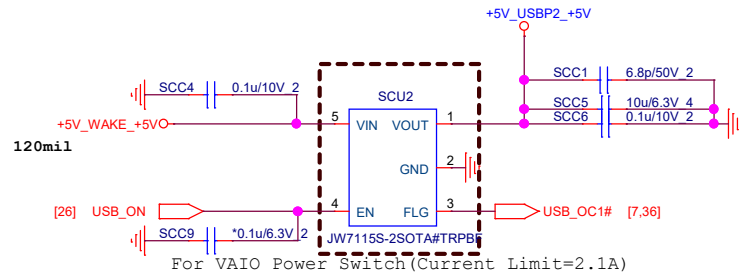
PD IT5271/EX Port2

This part should not contain any substances which are specified in EM-S303.

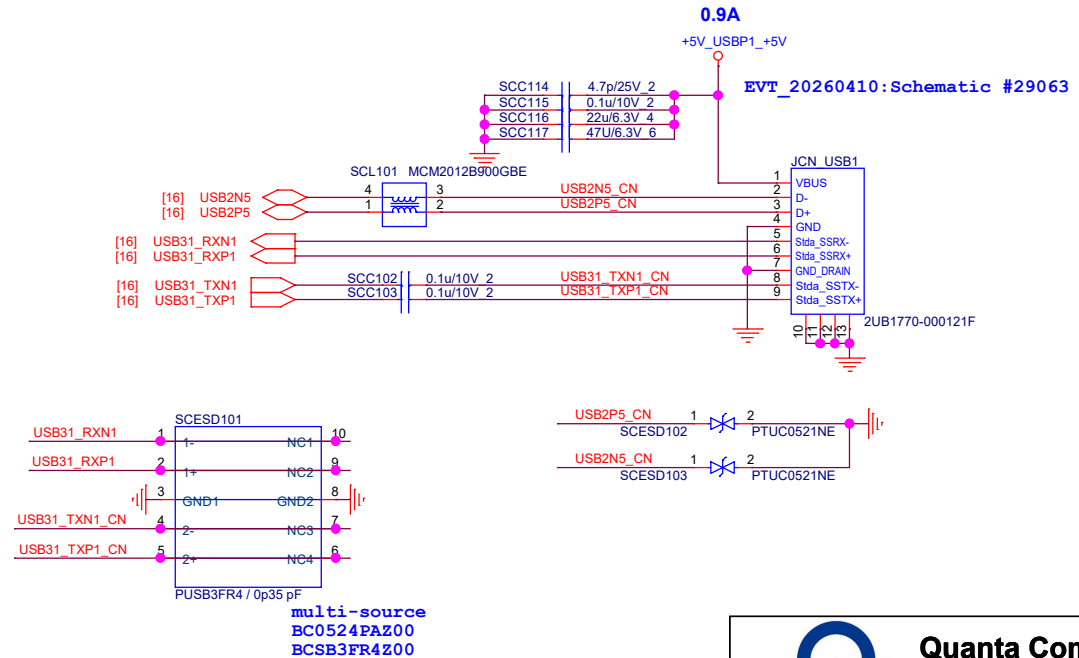
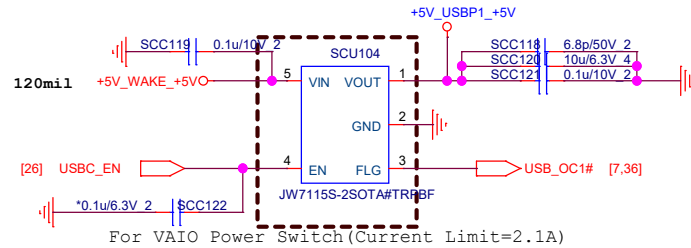


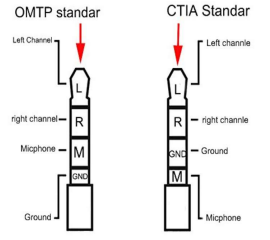


USB 3.2 GEN1 Type A Port2

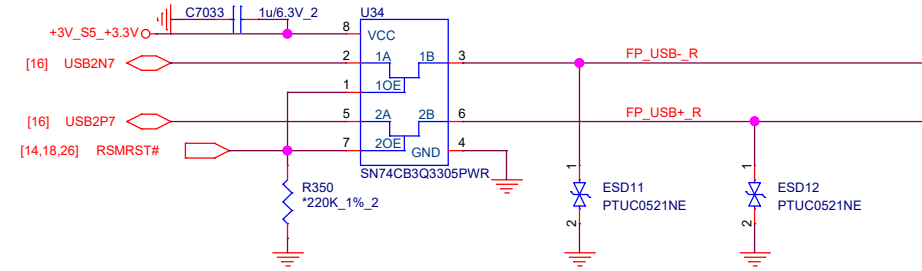
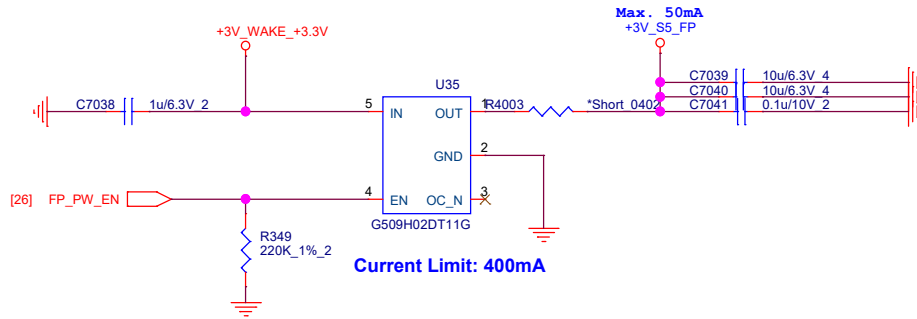


USB 3.2 GEN1 Type A Port1 (Near HDMI side)

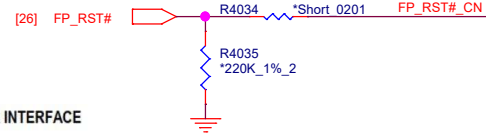




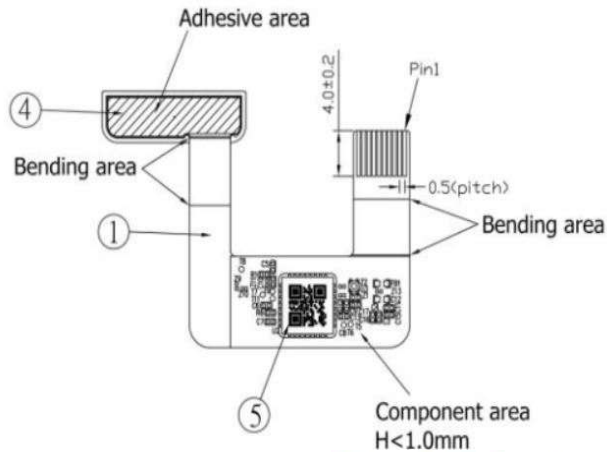
Fingerprint Power/BUS SW



EVT_20260331:Schematic #28974



Bottom view



2.2 USER INTERFACE

Pin Assignments

- Pin 1 = E_GND
 Pin 2 = USB_D+
 Pin 3 = USB_D-
 Pin 4 = GND
 Pin 5 = EC_STATE
 Pin 6 = EC_CTRL
 Pin 7 = (NC)
 Pin 8 = RST_N
 Pin 9 = FP_PRS#
 Pin 10 = VBUS(3V3)

Pull-up & Pull-down Resistors

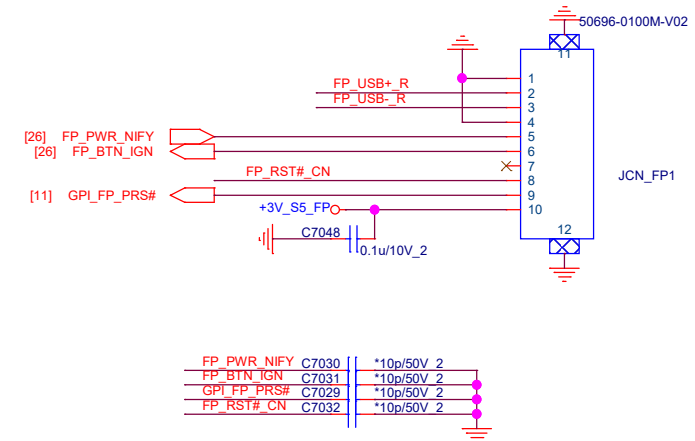
FP_PRS# (FP presence pin)	
High	Non Fingerprint SKU
Low	Fingerprint SKU

Symbol	Parameter	Minimum	Typical	Maximum	Unit
RPU	Pull-up resistor on RST_N input	34	-	100	kΩ
RPD	Pull-down resistor on TEST_CLK input	30	-	107	kΩ
RPDOTP	Pull-down resistor on VDDOTP	128	256	512	kΩ

MB Pull UP 100K , Fingerprint Module pull down 10K

This part should not contain any substances which are specified in EM-S303.

Fingerprint Module



Quanta Computer Inc.

PROJECT : FIN

Size	Document Number	Rev
	MB to FP	1A

Date: Monday, May 04, 2026 Sheet 38 of 59

[illegible]

The schematic diagram illustrates the LED driver circuit. A MOSFET (KBQ01 PDTA124EU) is used to switch the LED (KBESD2 PT05CFC) through a resistor (KBR28 33.5%_2). The MOSFET gate is driven by a signal [26] CAPS_LED# through a resistor (R4095 *0_5%_2). The LED is connected to a signal [18,25,26] LED_CATTERR#.

Vgs 0.5~1.0V
multi-source
BAM84380001
BAM10120003

BATTERY LED

BATTERY LED

Pin 1 of the LTST-C190KFKT LED is connected to the battery symbol.

Pin 2 of the LTST-C190KFKT LED is connected to the BATLED signal line.

The BATLED signal line is connected to pin 2 of the Q115 PDA124EU transistor.

The emitter of the Q115 PDA124EU transistor is connected to ground.

The collector of the Q115 PDA124EU transistor is connected to the +3VPCU_KBC_+3.3V supply through the R335 150_1%_2 resistor.

Two circuit diagrams for the LED driver section of the LTST-C1907GKT. The left diagram shows the connection for PWRLED1# to Q16 (PDTA124EU) and R336 (150_1%_2). The right diagram shows the connection for PWRLED2# to Q7023 (PDTA124EU) and R4086 (150_1%_2). Both diagrams include a +3VPCU_KBC_+3.3V supply and a DPWR2 1 input.

620mA

+5V_WAKE+5V

F2

SMF50603P100/1.0A/32V_0603

C7077
1u6.3V_2

R346
4.7K 5% 2

Q18
PJA3413

R347
4.7K 5% 2

620mA

VCC5_KBBL

C7081
0.1u/10V 2

C7082
0.1u/10V 2

PJE8438
Q7050

KB_BACKLIGHT [26]

R348
220K 1% 2

Vgs 0.5~1.0V
multi-source
BAM84380001
BAM10120003

[26] KBBL_PRESENT#

300mA

VCC5_KBBL

VCC5_KBBL

C7076 10pF/50V_2

C7078 1uF/6.3V_2

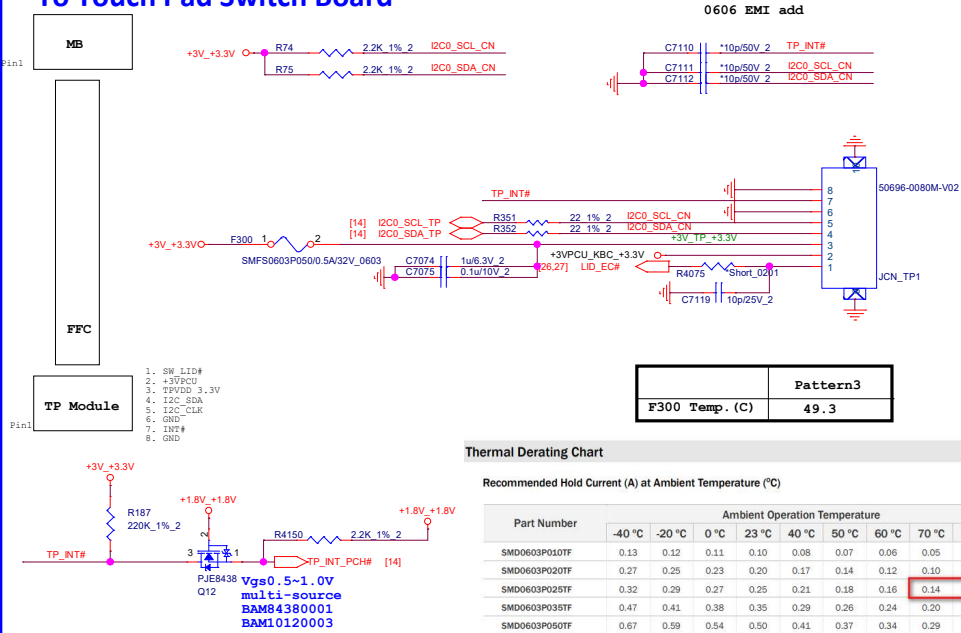
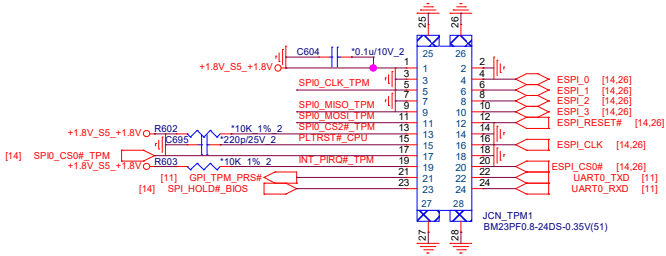
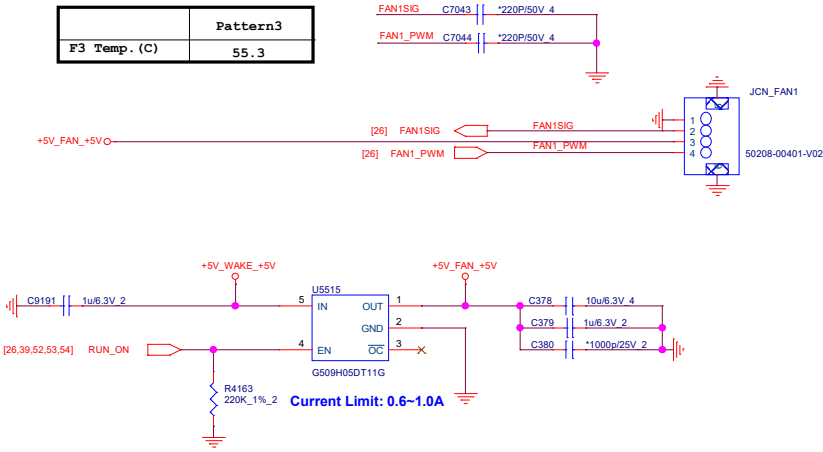
C7079 0.1uF/10V_2

JCN_KBBL1

196243-04041-3

	Pattern3
F2 Temp. (C)	

	Pattern3
F3 Temp. (C)	55.3

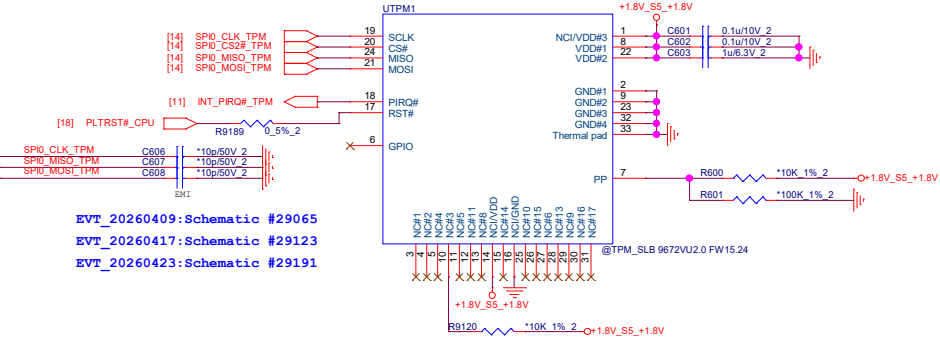


	Pattern3
F300 Temp. (C)	49.3

Thermal Derating Chart

Recommended Hold Current (A) at Ambient Temperature (°C)

Part Number	Ambient Operation Temperature							
	-40 °C	-20 °C	0 °C	23 °C	40 °C	50 °C	60 °C	85 °C
SMD0603P010TF	0.13	0.12	0.11	0.10	0.08	0.07	0.06	0.05
SMD0603P020TF	0.27	0.25	0.23	0.20	0.17	0.14	0.12	0.10
SMD0603P025TF	0.32	0.29	0.27	0.25	0.21	0.18	0.16	0.14
SMD0603P035TF	0.47	0.41	0.38	0.35	0.29	0.26	0.24	0.20
SMD0603P050TF	0.67	0.59	0.54	0.50	0.41	0.37	0.34	0.29



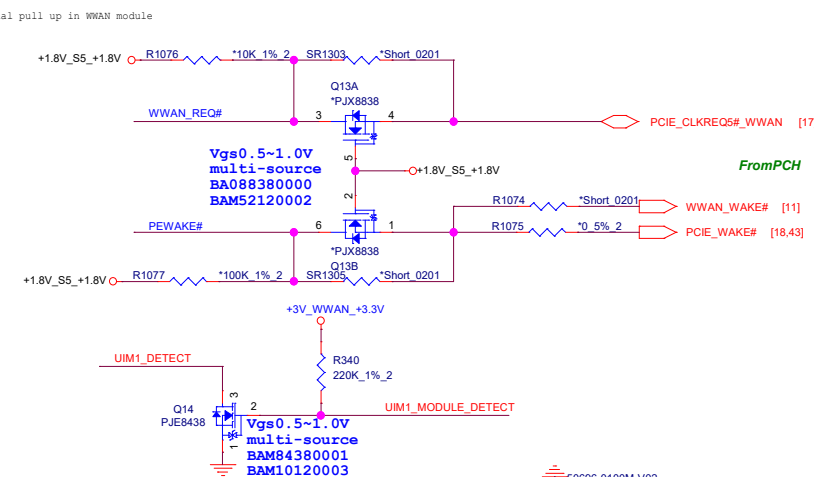
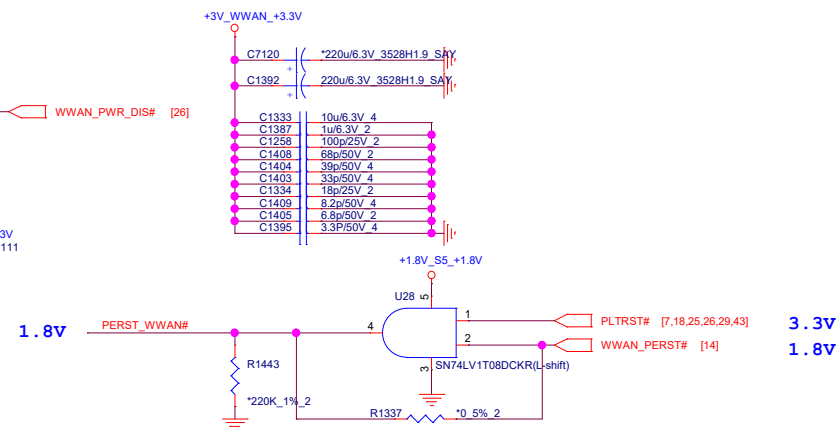
EVT_20260409:Schematic #29065
EVT_20260417:Schematic #29123
EVT_20260423:Schematic #29191

	W/ dtpm SKU	W/O dtpm SKU
UTPM1 (TPM IC)	O (Stuff)	X (Un-stuff)
R1372 (PD R)	O (Stuff)	X (Un-stuff)
R199 (PU R)	X (Un-stuff)	O (Stuff)



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Q7027A
PJK8838

CONFIG_1 3 4 CONFIG_1_PCH [11]

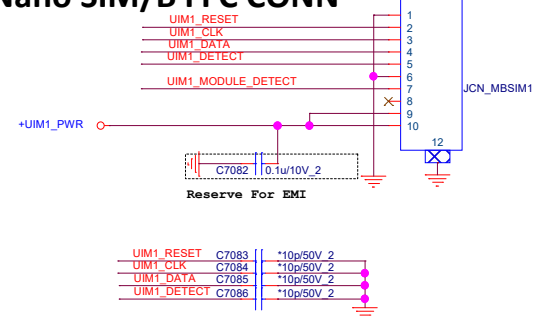
Vgs0_5-1.0V
multi-source
BA088380000
BAM52120002

+1.8V_SS_+1.8V

Q7027B
PJK8838

CONFIG_2 6 1 CONFIG_2_PCH [11]

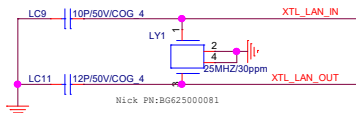
SIM Board	SIM	USIM1_DETECT_R	USIM1_MODULE_DETECT	USIM1_DETECT	Status
Exist	Exist	1	0	1	SIM
	Not-exist	0	0	0	no SIM
Not-exist	Exist	X	1	0	no SIM
	Not-exist	X	1	0	no SIM



LAN I219-LM/RTL8111K

Intel WGI219LM PHY Controller

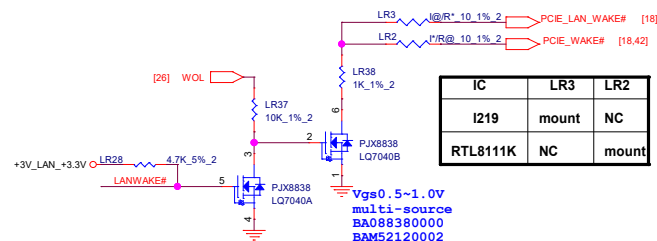
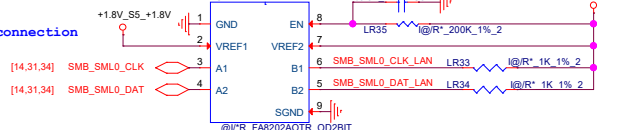
Vendor	Part Number	Description
INTEL	AL0SLKJ2005	IC (48P) WGI219LM-SLKJ2(QFN)VIOBS
REALTEK	AL008111016	IC CTRL(48P) RTL8111K-CG (QFN)



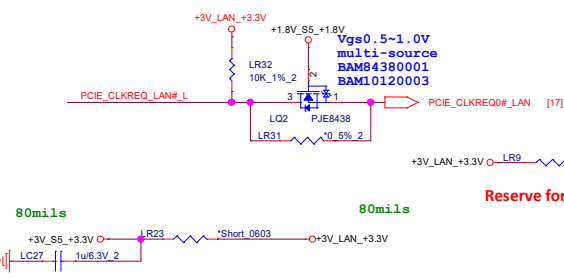
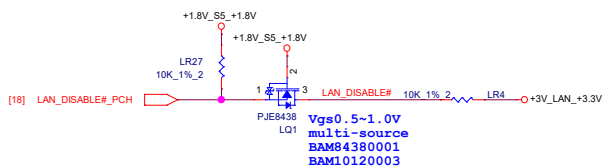
IC	LU4	LR33	LR34	LR35
I219	mount	mount	mount	mount
RTL8111K	NC	NC	NC	NC

EVT 20260410:Update PCH SM Link connection

EVT 20260410:Update PCH SM Link connection



IC	LR3	LR2
I219	mount	NC
RTL8111K	NC	mount

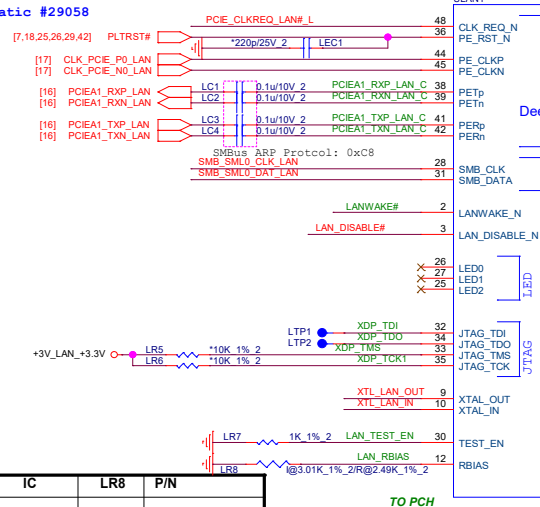


Reserve for LAN SW V-offset circuit.

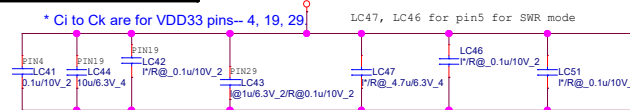
EVT 20260410:Update LAN schematic for i219/RTK8111K co-layout

EVT 20260413:Schematic #29058

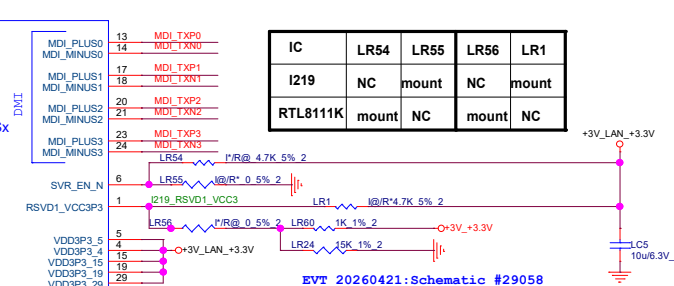
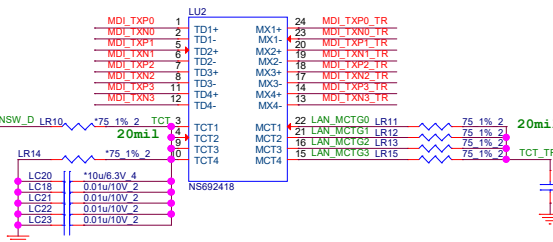
EVT 20260415:Schematic #29058



IC	LR8	P/N
I219	3.01K	CS23011FE04
RTL8111K	2.49K	CS22491FE02

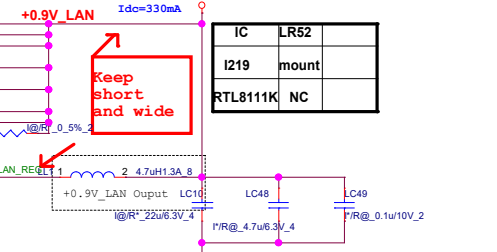


IC	LC42	LC43	P/N	LC47	LC46	LC51	LC52
I219	NC	1U	CH5101K9E05	NC	NC	NC	NC
RTL8111K	mount	0.1U	CH4102K9E00	mount	mount	mount	mount

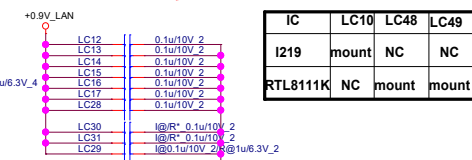


IC	LR54	LR55	LR56	LR1
I219	NC	mount	NC	mount
RTL8111K	mount	NC	mount	NC

EVT 20260421:Schematic #29058

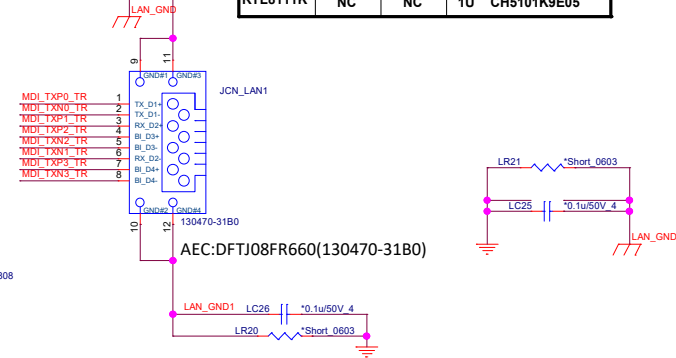


IC	LR52	
I219	mount	
RTL8111K	NC	



IC	LC10	LC48	LC49
I219	mount	NC	NC
RTL8111K	NC	mount	mount

IC	LC30	LC31	LC29 P/N
I219	mount	mount	0.1U CH4102K9E00
RTL8111K	NC	NC	1U CH5101K9E05

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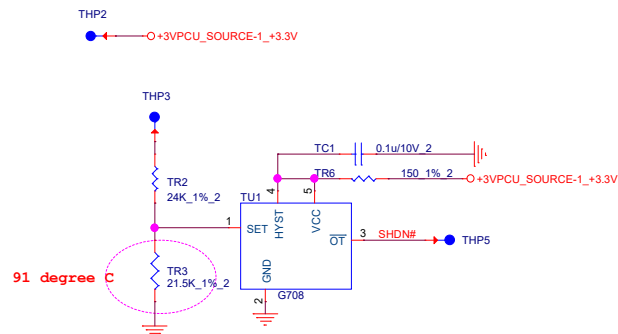
PROJECT : SPKZ

Size	Document Number	Re
LAN INTEL I219LM		
Date:	Monday, May 04, 2026	Sheet 43 of 57

This part should not contain any substances which are specified in EM-S303

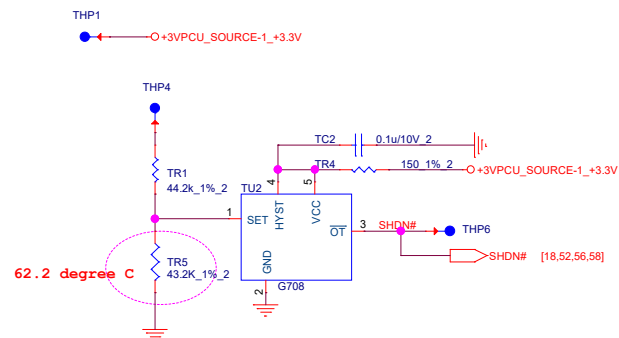
H/W Thermal Protect

As close as possible to CPU


$$R_{SET}(k\Omega) = 0.0012T^2 - 0.9308T + 96.147$$

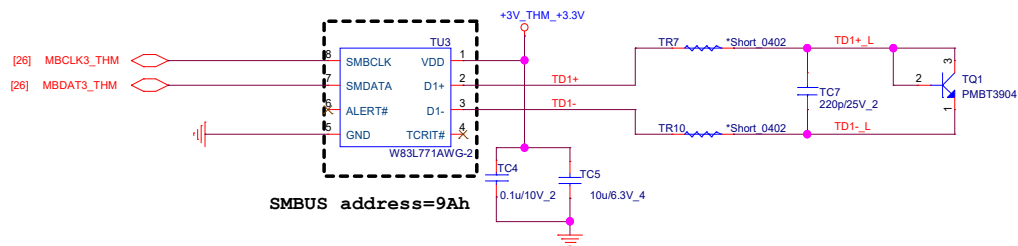
UMA SKU					
Location of IC	Temp	R-Set	Parts in BOM	Max	Min
Near USB3 CN sensor temp	62.2	TR5=42.8K	TR5=43.2K	62.4	61.1
Near CPU sensor temp	91	TR3=21.3K	TR3=21.5K	91.7	89.8

Thermally far from heat source

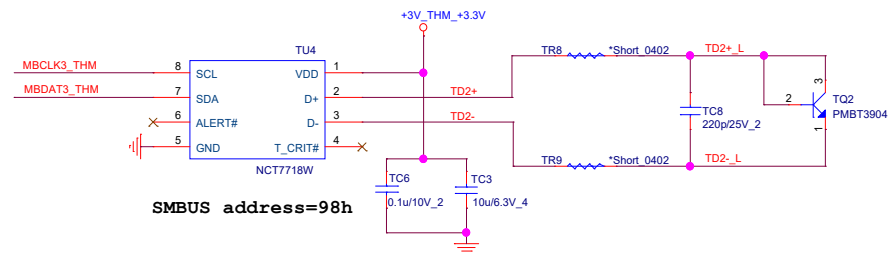


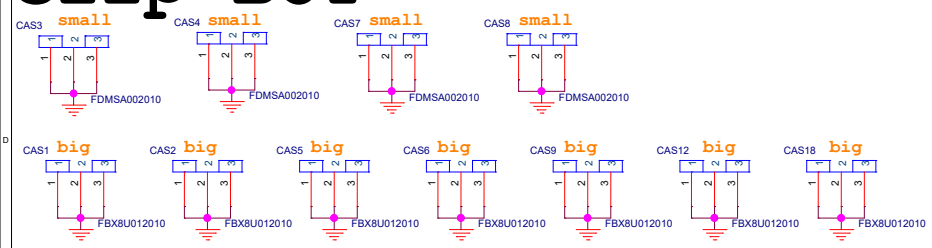
S/W Thermal Protect

Close to main heatpipe

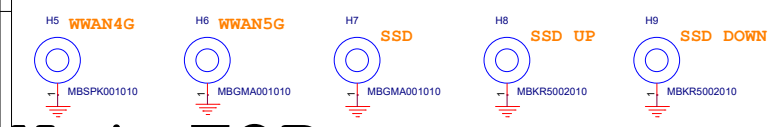


SMBUS address=98h





Nut BOT



Nut TOP

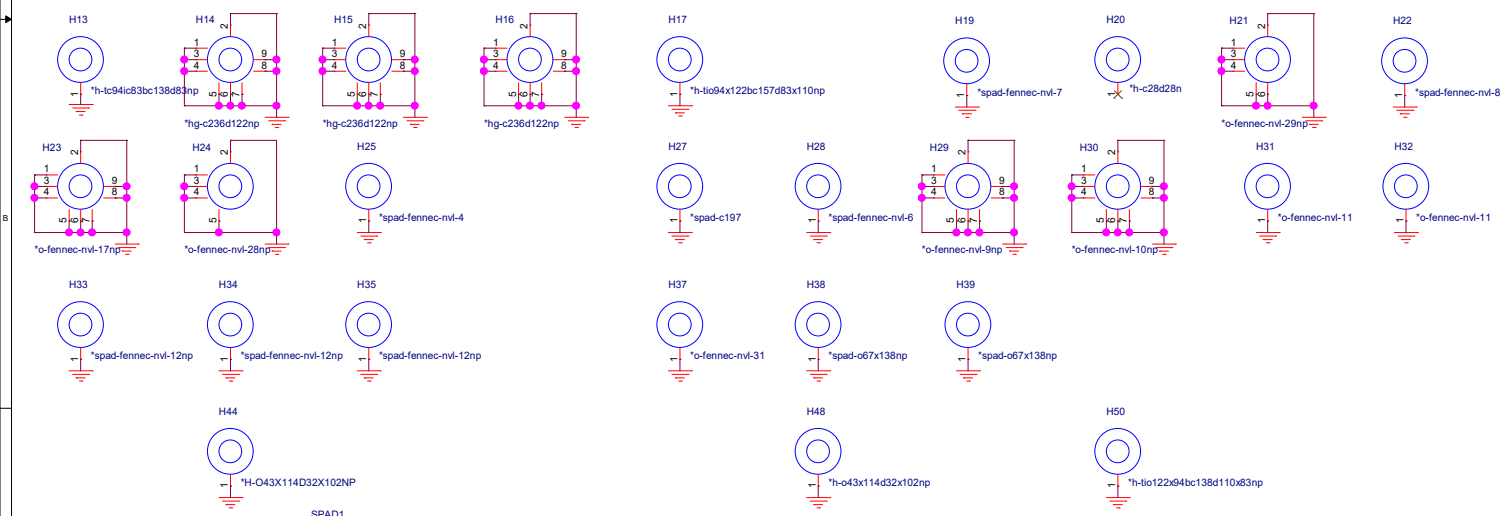


Hole

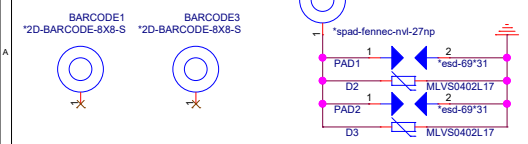
EVT_20260409: H45,H41,H12,H46 removed, CAS3 FP change to CLIP-79X79-93X24-3P

EVT_20260410: SPAD1,H23,H24,H10,H11,H1,H2,H3,H4,H50,H48 change footprint. Add H51,H52,H53,H54. Remove H42,H47,H18,H49,H26

EVT_20260417: CAS3,SPAD1,H24, change footprint. Remove CAS11



2D Barcode



EVT_20260427: Board data #29275

EVT_20260430: Remove H40, H43, H36

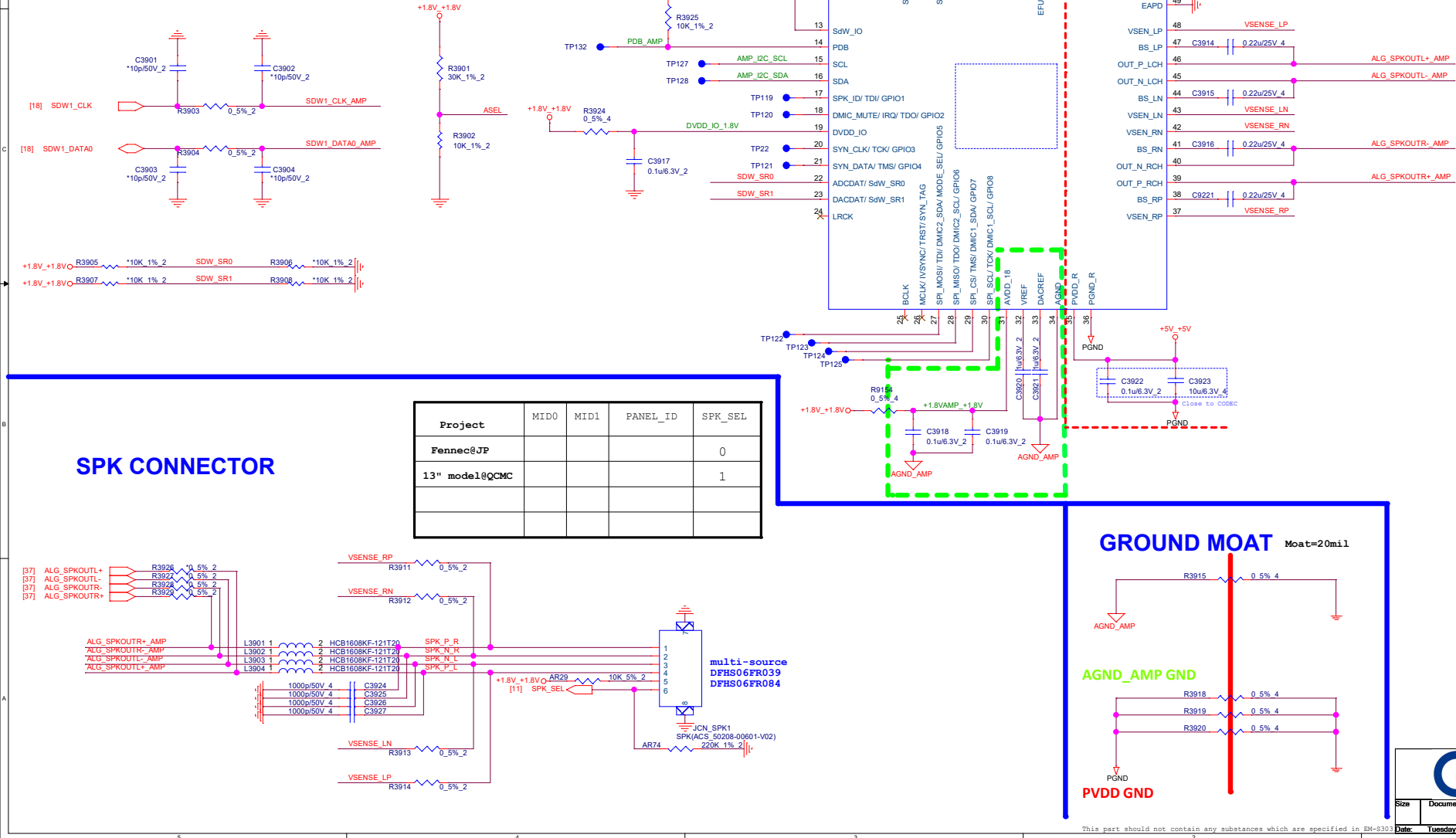


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SCREW HOLE		1A
Date: Monday, May 04, 2026	Sheet 45 of 59	

SdW Address ID selection

ASEL pin	Ra1/Ra2	Rb1/Rb2	Device ID
>1.65V	0	-	33_02_5d_13_20_00h
1.65V~1.17V	10K	30K	32_02_5d_13_20_00h
1.17V~0.63V	20K	20K	31_02_5d_13_20_00h
0.63V~0.21V	30K	10K	30_02_5d_13_20_00h



This part should not contain any substances which are specified in BM-6303

NVL H16C 12XE 25W
 PL1=25W, PL=65W, PL4=163W
MP29021/MP87440C
Schematic:H@

Phase	Power Rail	IPL2	ICCMAX	DC LL	AC LL	Freq
3	VCORE	44A	118A	4mohm	4mohm	500khz
2	VCCGT	40A	103A	0mohm	4mohm	500khz
2	VCCSA	39A	68A	4mohm	4mohm	500khz
1	VCCLP	15A	23A	5.8mohm	5.8mohm	500khz

NVL-H16C 4XE 25W
 PL1=25W, PL=65W, PL4=163W
MP29021/MP87440C
Schematic:E@

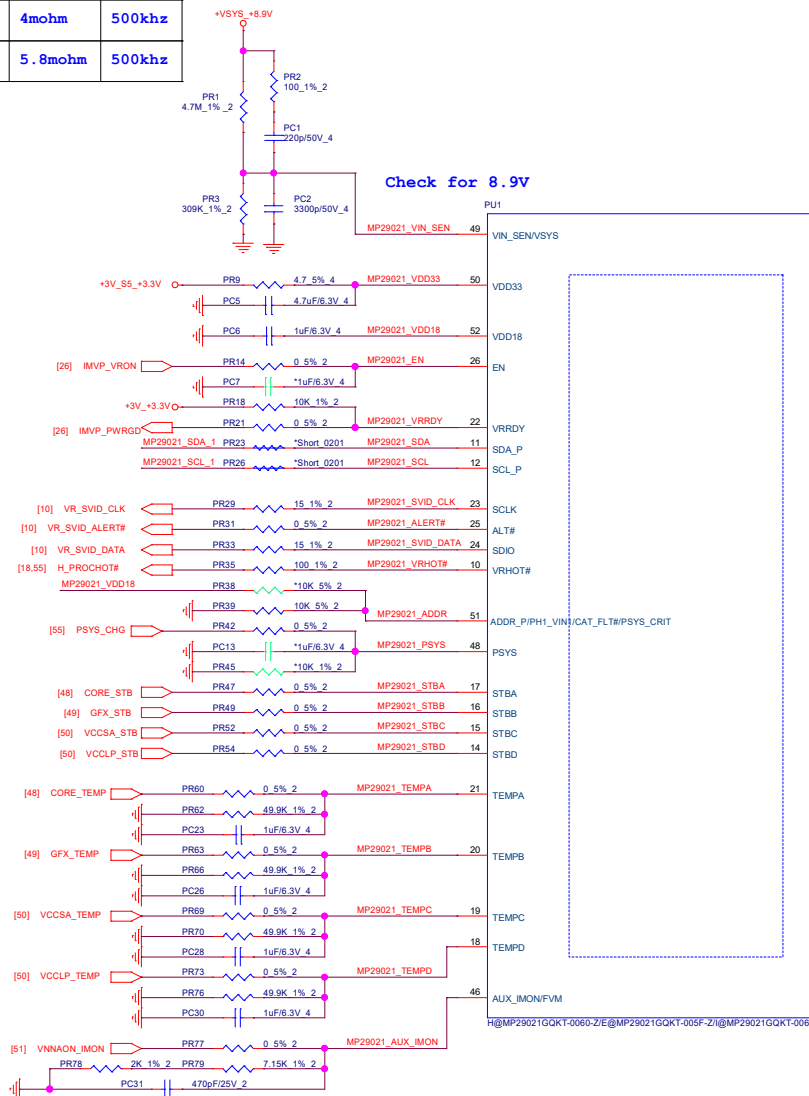
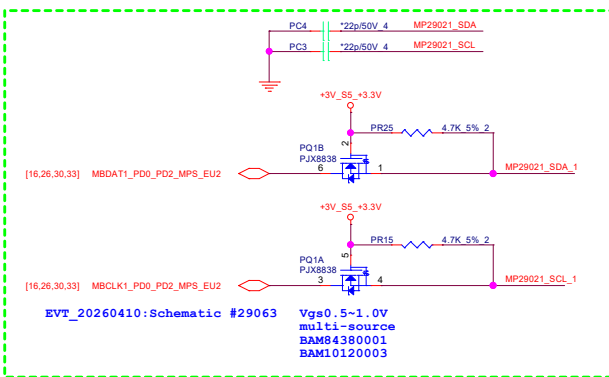
Phase	Power Rail	IPL2	ICCMAX	DC LL	AC LL	Freq
3	VCORE	44A	118A	4mohm	4mohm	500khz
1	VCCGT	23A	57A	0mohm	4mohm	500khz
2	VCCSA	39A	68A	4mohm	4mohm	500khz
1	VCCLP	15A	23A	5.8mohm	5.8mohm	500khz

MP29021 Setting:
 Switching Frequency Setting= 500 KHz
 APS(Auto Phase Shielding): Default enable

NVL H8C 15W
 PL1=15W, PL=45W, PL4=124W
MP29021/MP87440C
Schematic:I@

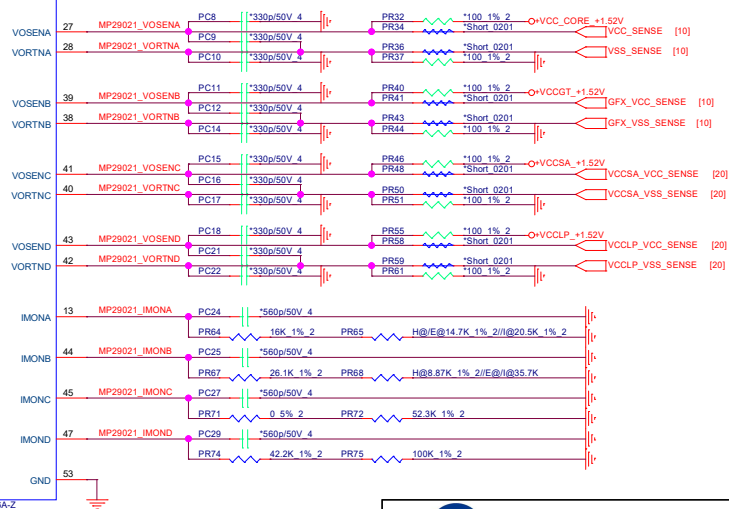
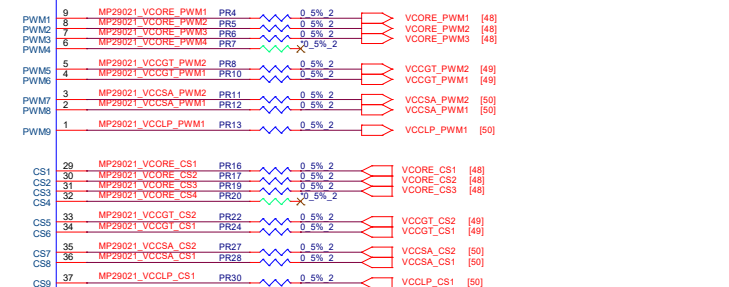
Phase	Power Rail	IPL2	ICCMAX	DC LL	AC LL	Freq
2	VCORE	33A	99A	4mohm	4mohm	500khz
1	VCCGT	23A	57A	0mohm	4mohm	500khz
2	VCCSA	39A	68A	4mohm	4mohm	500khz
1	VCCLP	15A	23A	5.8mohm	5.8mohm	500khz

CPU Type		PU1	PR65	PR68
H16C 12Xe	H@	AL029021002	CS31471FE01	CS28871FE01
H16C 4Xe	E@	AL029021004	CS31471FE01	CS33571FE02
H8C 4Xe	I@	AL029021003	CS32051FE00	CS33571FE02

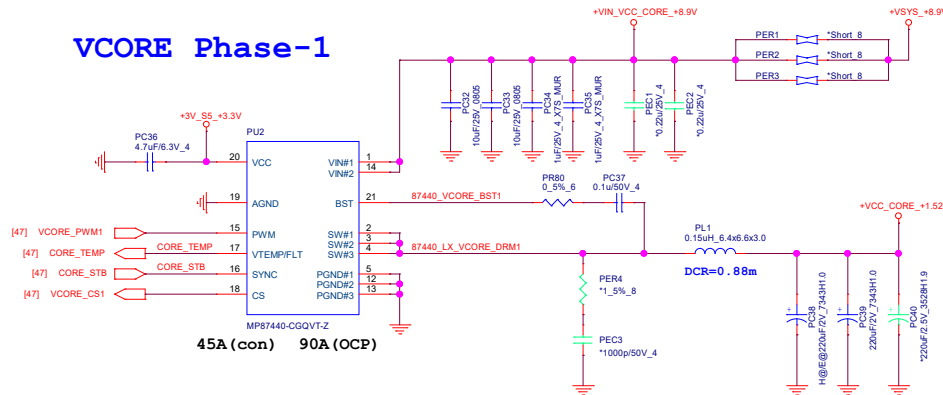


Register Configurable Phase:

Option	Rail 1	Rail 2	Rail 3	Rail 4
3+2+2+1	PH1~4	PH5~6	PH7~8	PH9



VCORE Phase-1



NVL-H12Xe (Performance):

VCORE IccMAX=118A, Itrip_nom=75A, TDC=44A
DC-LL=4m AC-LL=4m

OVP: 1.92V or Vout+400mV

NVL-H16C 4Xe (Performance):

VCORE IccMAX=118A, Itrip_nom=75A, TDC=44A
DC-LL=4m AC-LL=4m

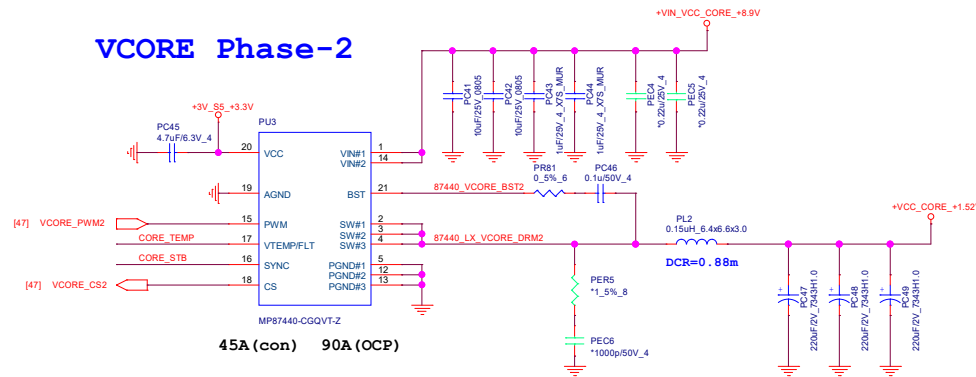
OVP: 1.92V or Vout+400mV

NVL-H8C (Performance):

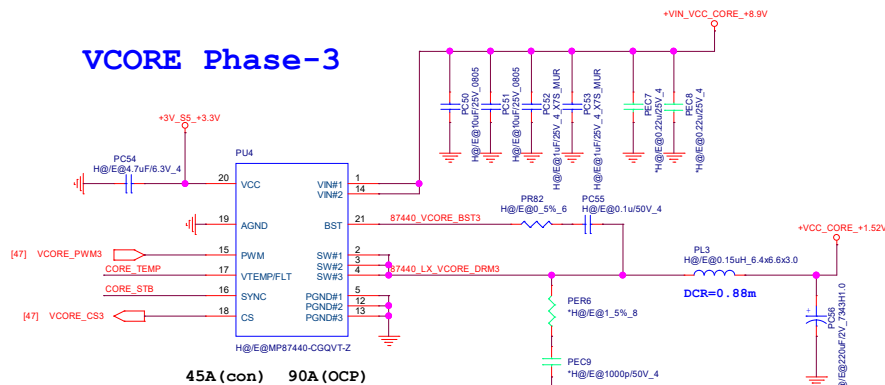
VCORE IccMAX=99A, Itrip_nom=75A, TDC=33A
DC-LL=4m AC-LL=4m

OVP: 1.92V or Vout+400mV

VCORE Phase-2



VCORE Phase-3

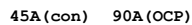


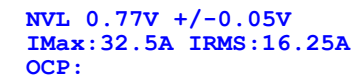
VCC GFX Phase1



NVL-H8C (Performance):
VCCGT IccMAX=57A, Itrip_nom=37A,TDC=23A
DC-LL=0 AC-LL=4m

VCC GFX Phase2 (12Xe only)



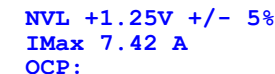


State	Fsw	Rmode
M1	700KHz	0
M2	900KHz	>230k or floating

State	Current Limit	CLM Resistor to GND
M1	10.5A	0
M2	14.5A	90.9k
M3	18.5A	150k
M4	21.5A	>230k or floating

VID	Reference (V)
0	0.6V
1	1.2V

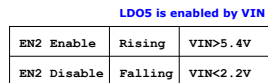
+VCCPRIM IO



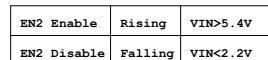
MIN. Current Limit	ILMT PIN
6A	LOW
8A	FLOATING
10A	HIGH

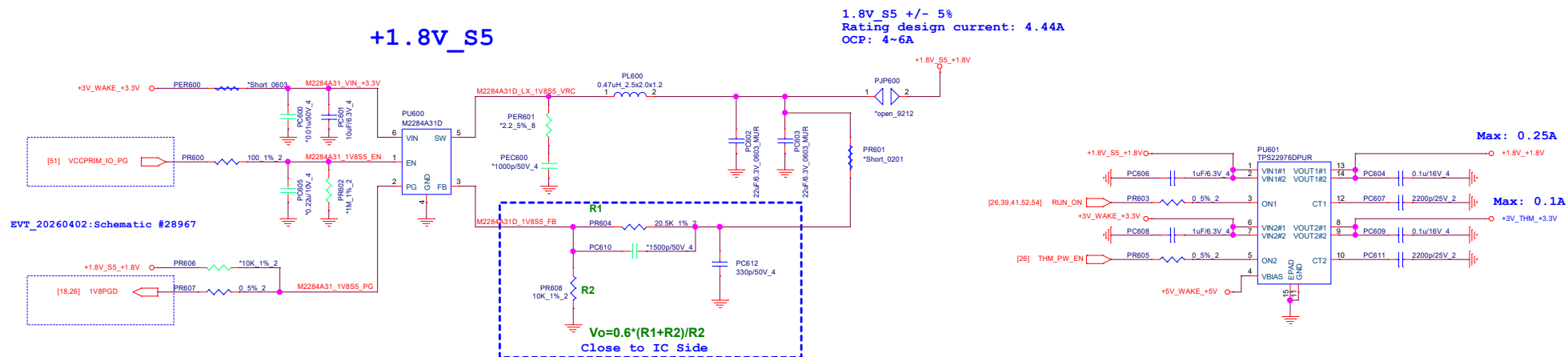
EN Voltage Level	PWM Mode
1.0V ~ 1.6V	Ultrasonic
2.0V ~ VIN	PFM

+5V WAKE



+3V WAKE



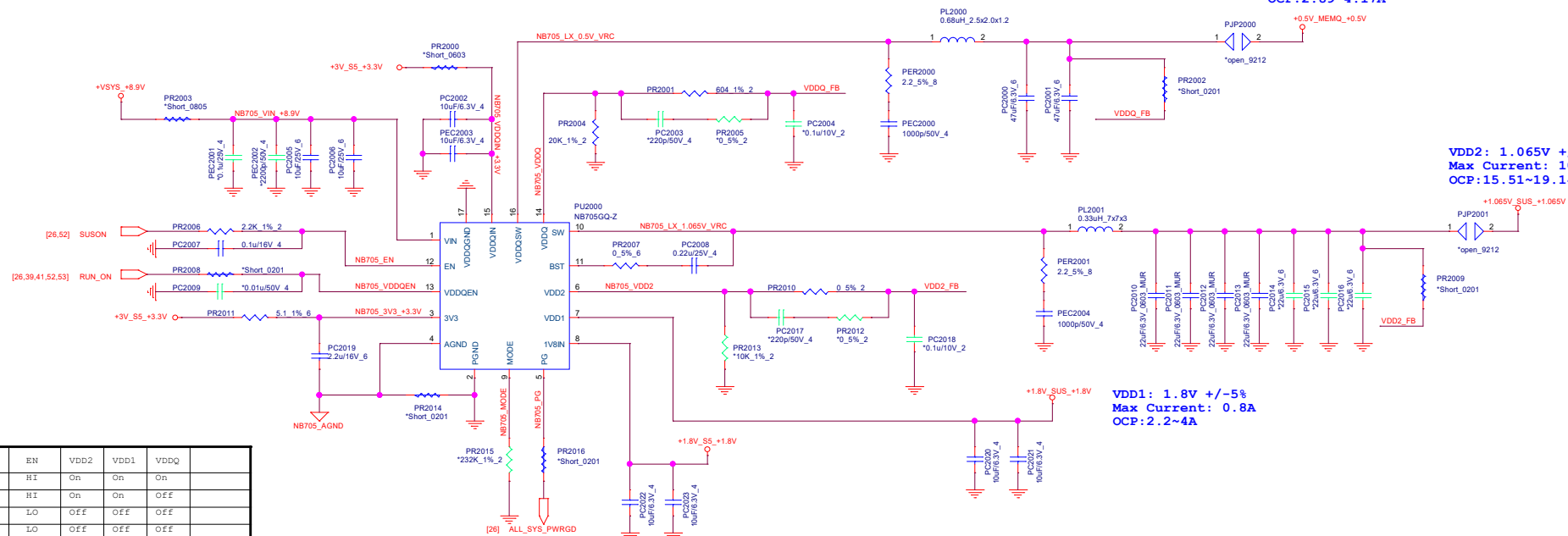


LPDDR5_PWR +1.065V_SUS & +1.8V_SUS & VDDQ

VDDQ: 0.52V +/-5%
Max Current: 2.35A
OCP: 2.89~4.17A

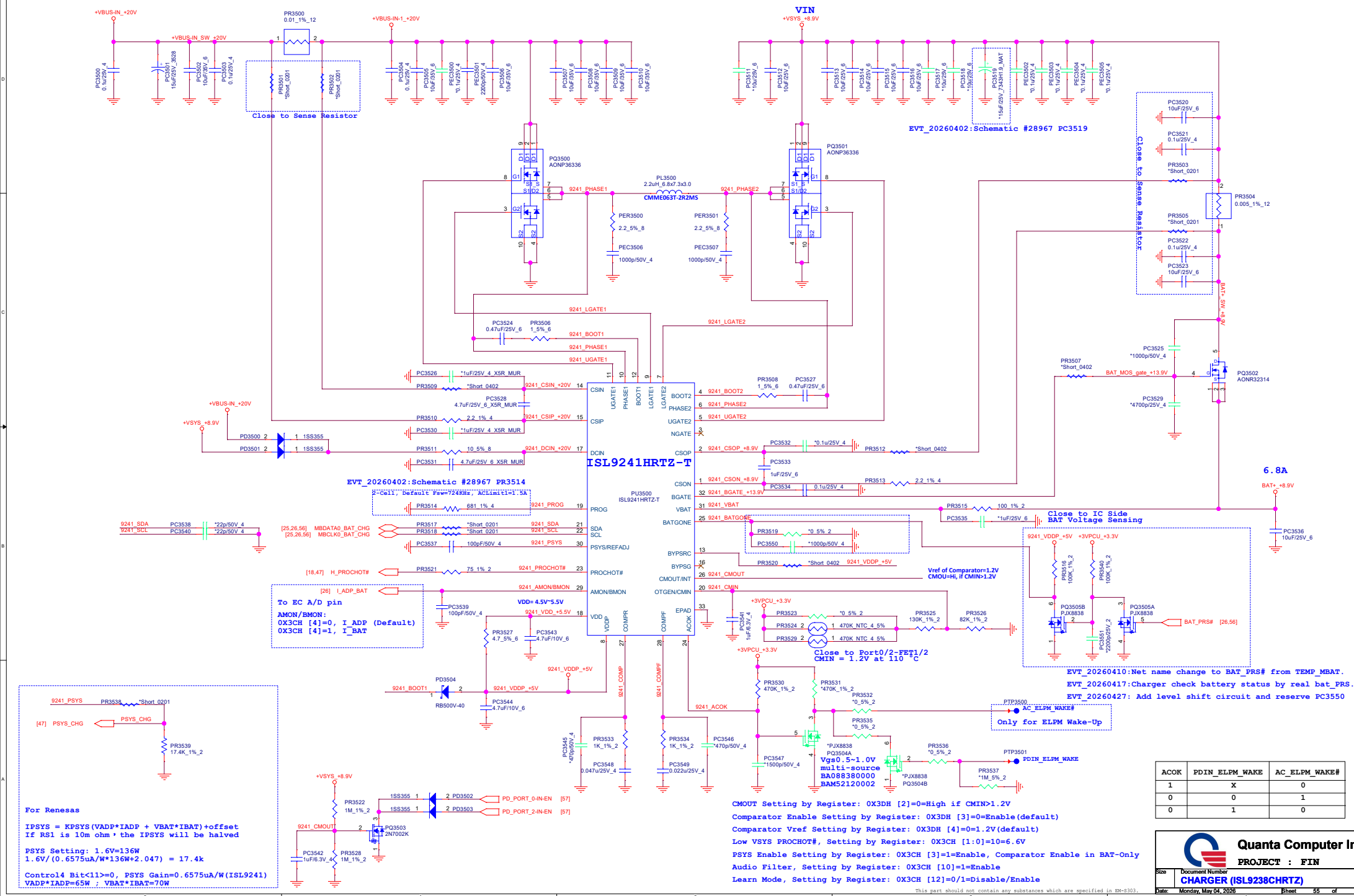
VDD2: 1.065V +/-5%
Max Current: 10.16A
OCP: 15.51~19.14A

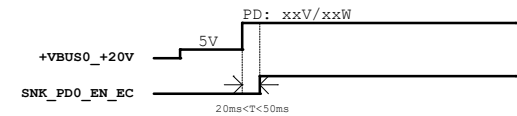
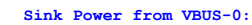
VDD1: 1.8V +/-5%
Max Current: 0.8A
OCP: 2.2~4A



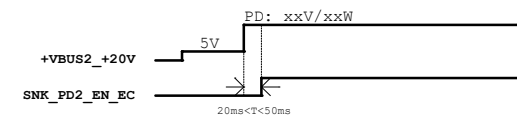
VDDQEN	EN	VDD2	VDD1	VDDQ	
HI	HI	On	On	On	
LO	HI	On	On	Off	
LO	LO	Off	Off	Off	
HI	LO	Off	Off	Off	

Fsw=700KHz

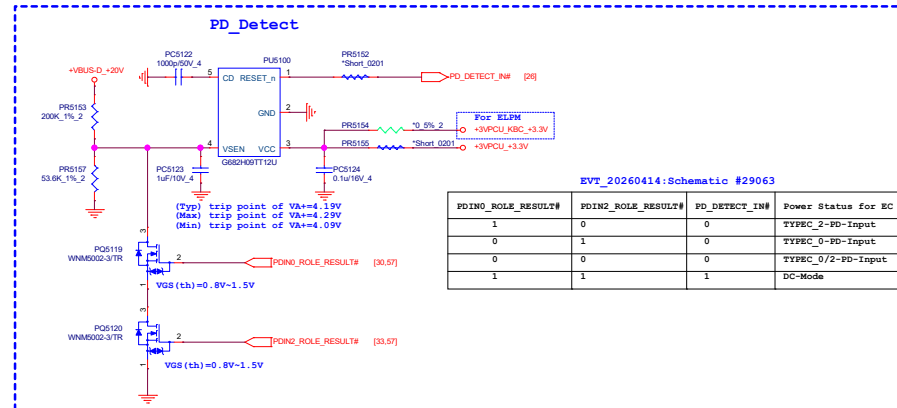
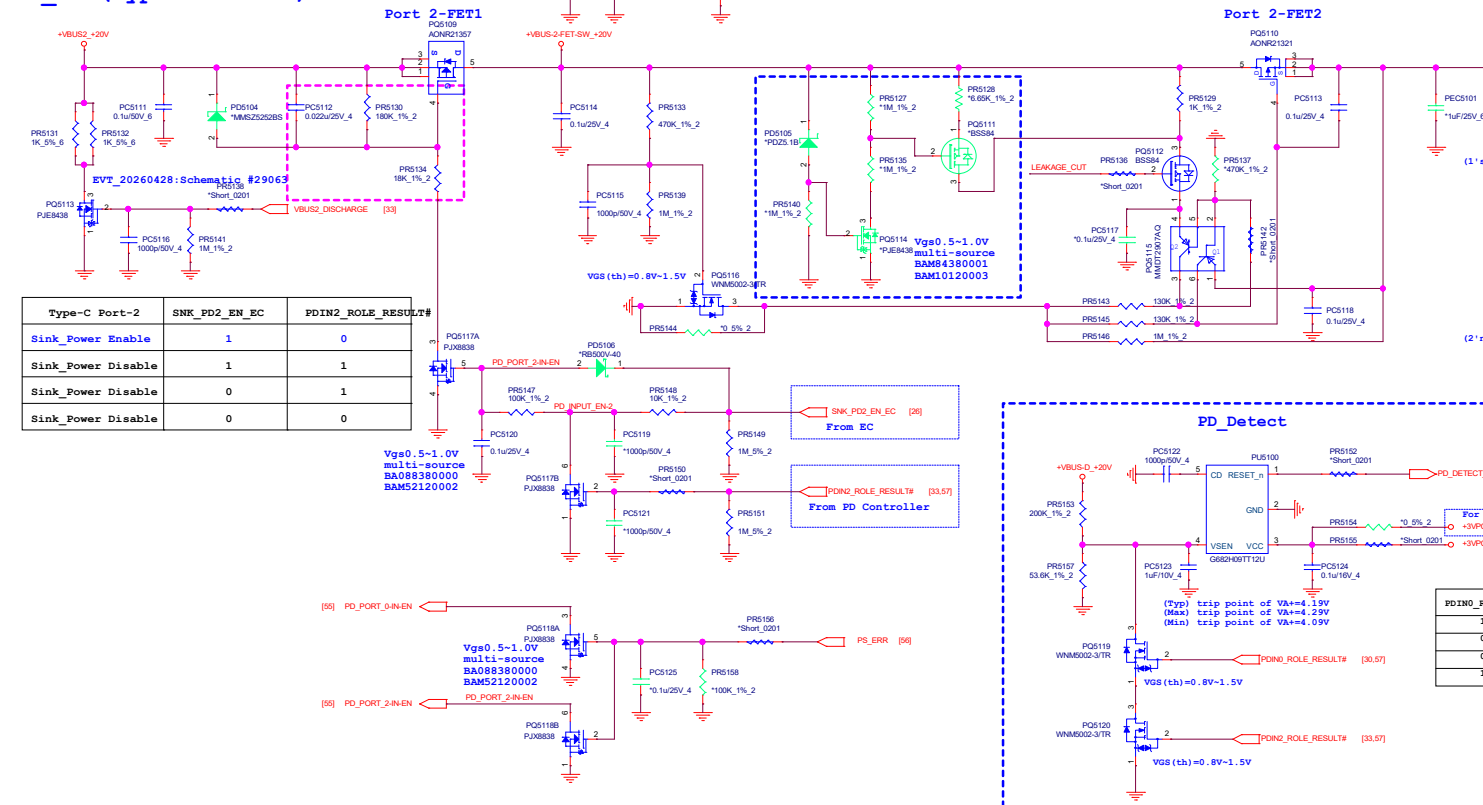
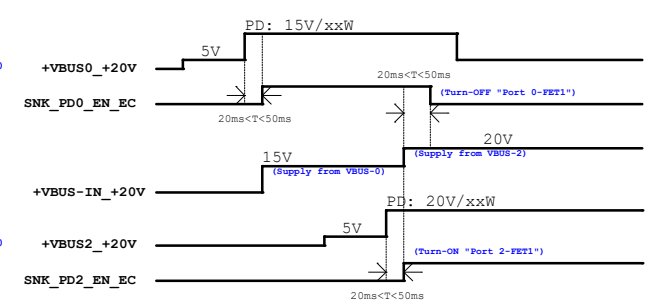




Sink Power from VBUS-2:

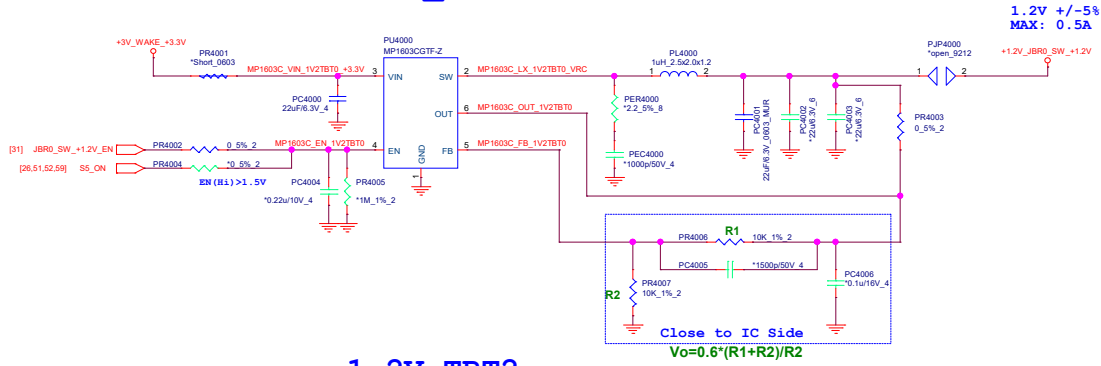


Sink Power Transition (From VBUS-0 to VBUS-2):

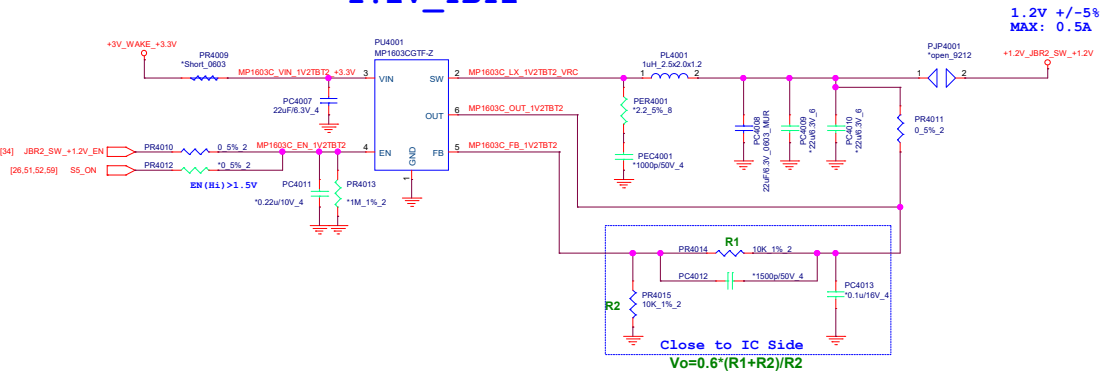


EVT_20260414:Schematic #29063			
PDINO_ROLE_RESULT#	PDIN2_ROLE_RESULT#	PD_DETECT_IN#	Power Status for
1	0	0	TYPEC_2-PD-Input
0	1	0	TYPEC_0-PD-Input
0	0	0	TYPEC_0/2-PD-Input
1	1	1	DC-Mode

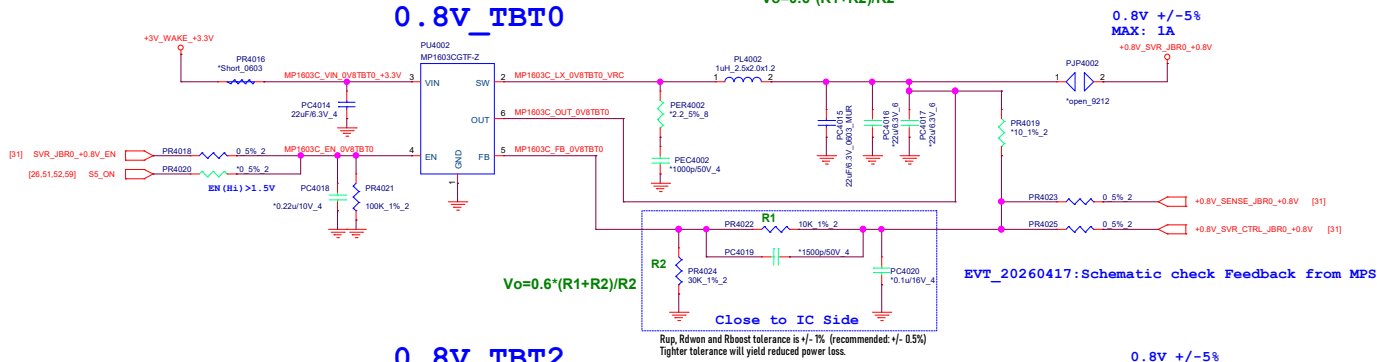
1.2V_TBT0



1.2V_TBT2



0.8V_TBT0



0.8V_TBT2

